

EXERCISES

LEVEL I

(Questions Based on Fundamentals)

nth TERM OF AN AP

- In an AP, if its $t_n = 3n + 5$, find its common difference.
- If the p th term of an AP is q and the q th term is p , prove that its n th term is $(p + q - n)$.
- In an AP, if $mt_m = nt_n$, show that $t_{m+n} = 0$.
- If the m th term of an AP be $\frac{1}{n}$ and n th term be $\frac{1}{m}$, then show that its (mn) th term is 1.
- If $(m + 1)$ th term of an AP is twice the $(n + 1)$ th term, prove that $(3m + 1)$ th term is twice the $(m + n + 1)$ th term.
In an AP, if $T_p = q$, $T_{p+q} = 0$, find T_q .
- A man starts repaying a loan as first instalment of ₹100.
- If he increases the instalment by ₹5 every month, what amount he will pay in the 30th instalment?
- In an AP, prove that $t_{m+n} + t_{m-n} = 2t_m$.
- If p th, q th, r th terms of an AP be a , b , c respectively, show that:
(i) $a(q - r) + b(r - p) + c(p - q) = 0$
(ii) $(a - b)r + (b - c)p + (c - a)q = 0$
- If $\langle a_n \rangle$ is an AP such that $\frac{a_4}{a_7} = \frac{2}{3}$, find $\frac{a_6}{a_8}$.

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SELECTIONS OF TERMS IN AN AP

- If a, b, c, d, e are in AP, find the value of $a - 4b + 6c - 4d + e$.
- Divide 69 into three parts to form an AP and the product of the two smaller parts is 483.
The angles of a quadrilateral are in AP whose common difference is 10° . Find the angles.
- If the roots of the equation $x^3 - 12x^2 + 39x - 28 = 0$ are in AP, find the common difference.

- Divide 20 into four parts to form an AP such that the ratio of the product of the first and the fourth to the product of the second and third is $2 : 3$.

- If $a_1, a_2, \dots, a_n$ are in AP, prove that

$$\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \dots + \frac{1}{a_{n-1} a_n} = \frac{n-1}{a_1 a_n}$$

- If $a_1, a_2, \dots, a_n$ are in AP, prove that

$$\frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}} = \frac{n-1}{\sqrt{a_1} + \sqrt{a_n}}$$

$$a_i > 0 \text{ for all } i$$

- If $a_1, a_2, \dots, a_n$ are in AP, prove that

$$\frac{1}{a_1 a_n} + \frac{1}{a_2 a_{n-1}} + \frac{1}{a_3 a_{n-2}} + \dots + \frac{1}{a_n a_1} = \frac{2}{a_1 + a_n} \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right)$$

- In an AP, $t_7 = 15$, find the value of common difference, d that would make the value of t_2, t_7, t_{12} maximum.
- If the non-zero numbers a, b, c are in AP and $\tan^{-1} a, \tan^{-1} b, \tan^{-1} c$ are also in AP, prove that
(i) $b^2 = ac$
(ii) $a = b = c$

SUM TO N -TERMS OF AN AP

- Find the sum of the series $5 + 13 + 21 + \dots + 181$.
- Find the sum of all three-digit natural numbers which are divisible by 7.
- Find the sum of all natural numbers between 250 and 1000 which are exactly divisible by 3.
- Find the sum of first 24 terms of the AP $a_1, a_2, a_3, \dots, a_n$ if it is known that $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$.

25. In an AP,
 (i) if $S_n = 3n^2 + 4n$, find its 10th term
 (ii) if $S_n = 2n^2 + 5n$, find its common difference.
26. If the m th term of an AP is $\frac{1}{n}$ and the n th term is $\frac{1}{m}$,
 show that the sum of m th term is $\frac{1}{2}(mn + 1)$.
27. Solve for x $1 + 6 + 11 + 16 + \dots + x = 148$.
28. The sum of the first p, q, r terms of an AP are a, b, c respectively. Show that

$$\frac{a}{p}(q-r) + \frac{b}{q}(r-p) + \frac{c}{r}(p-q) = 0$$

29. The sum of $n, 2n, 3n$ terms of an AP are S_1, S_2, S_3 respectively. Prove that

$$S_3 = 3(S_2 - S_1).$$

30. The p th term of an AP is a and q th term is b . Prove that the sum of its $(p+q)$ terms is

$$\left(\frac{p+q}{2}\right)\left\{a+b+\frac{a-b}{p-q}\right\}.$$

31. If in an AP, the sum of m terms is equal to n and the sum of n terms is equal to m , prove that the sum of $(m+n)$ terms is $-(m+n)$.

32. The ratio of the sum of n terms of two APs is $\frac{(7n+1)}{(4n+27)}$.
 Find the ratio of their 11th terms.

33. The ratio of the sum of m and n terms of an AP is $m^2 : n^2$. Show that the ratio of the m th and n th terms is $(2m-1) : (2n-1)$.

34. The interior angles of a polygon are in AP. The smallest angle is 120° and the common difference is 5° . Find the number of sides of the polygon.

35. If $S_1, S_2, S_3, \dots, S_m$ are the sum of n terms of m APs whose first terms are $1, 2, 3, \dots, m$ and the common differences are $1, 3, 5, \dots, (2m-1)$, respectively. Show that

$$S_1 + S_2 + \dots + S_m = \frac{mn}{2}(mn+1).$$

36. If $S_n = n^2p$ and $S_m = m^2p$, $m \neq n$, in an AP, prove that $S_p = p^3$.

37. If S_1 be the sum of $(2n+1)$ terms of an AP and S_2 be the sum of its odd terms, prove that

$$S_1 : S_2 = (2n+1) : (n+1).$$

38. If S_1 be the sum of odd terms and S_2 be the sum of even terms of an AP consists of $(2n+1)$ terms, prove that $S_1 : S_2 = (n+1) : n$.

39. If the sum of n terms of an AP is $nP + \frac{1}{2}n(n-1)Q$,

where P and Q are constants, find the common difference.

PROPERTIES OF AP

40. If a, b, c are in AP, prove that the following are also in AP.

(i) $b+c, c+a, a+b$

(ii) $a^2(b+c), b^2(c+a), c^2(a+b)$

(iii) $\frac{1}{\sqrt{b}+\sqrt{c}}, \frac{1}{\sqrt{c}+\sqrt{a}}, \frac{1}{\sqrt{a}+\sqrt{b}}$

(iv) $\frac{1}{bc}, \frac{1}{ca}, \frac{1}{ab}$

(v) $a\left(\frac{1}{b}+\frac{1}{c}\right), b\left(\frac{1}{c}+\frac{1}{a}\right), c\left(\frac{1}{a}+\frac{1}{b}\right)$

(vi) $[(b+c)^2 - a^2], [(a+c)^2 - b^2], [(b+a)^2 - c^2]$

41. If a^2, b^2, c^2 are in AP, prove that the following terms are also in AP.

(i) $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$

(ii) $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$

42. If a, b, c are in AP, show that

(i) $a^2(b+c), b^2(c+a), c^2(a+b)$ are in AP.

(ii) $b+c-a, c+a-b, a+b-c$ are in AP.

(iii) $bc-a^2, ca-b^2, ab-c^2$ are in AP.

43. If $\frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ are in AP, prove that

$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are also in AP.

INSERTION OF AM

44. Insert three arithmetic means between 3 and 19.

45. For what value of n , $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}$ is the arithmetic mean of a and b ?

46. If n arithmetic means are inserted between 20 and 80 such that the ratio of first mean to the last mean is $1:3$, find the value of n .

47. Prove that the sum of n arithmetic means between two numbers is n times the single AM between them.

48. If the AM between p th and q th terms of an AP be equal to the AM between r th and s th terms of the AP, show that $p+q=r+s$.

49. There are n AMs between 3 and 17. If the ratio of the last mean to the first mean is $3:1$, find the value of n .

50. If x, y, z are in AP and A_1 is the AM of x and y ; and A_2 is the AM of y and z , prove that the AM of A_1 and A_2 is y .

GEOMETRIC PROGRESSION

n th term of the GP

51. Find the 10th term of the GP $\sqrt{2}, \frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}, \dots$

52. Which term of the GP $\sqrt{2}, \frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}, \frac{1}{4\sqrt{2}}, \dots$ is $\frac{1}{512\sqrt{2}}$?

53. Which term of the GP $2, 1, \frac{1}{2}, \frac{1}{4}, \dots$ is $\frac{1}{128}$?
54. If the 2nd and 5th terms of a GP are 24 and 81, respectively, find the GP.
55. If the 3rd term of a GP is 3, then find the product of its first five terms.
56. The sum of three numbers in GP is 21 and the sum of their squares is 189, find the numbers.
57. If the first term of a GP is 1 and the sum of the 5th and 1st term is 82, find the common ratio.
58. If the p th, q th and r th terms of a GP are a, b and c , respectively, prove that

$$a^{q-r} \cdot b^{r-p} \cdot c^{p-q} = 1$$
59. If the first and the n th terms of a GP are a and b , respectively and if P is the product of the first n terms, prove that $P^2 = (ab)^n$.
60. The $(m+n)$ th and $(m-n)$ th terms of a GP are p and q , respectively. Show that the m th and n th terms are $\sqrt{pq}$ and $p\left(\frac{q}{p}\right)^{m/2n}$, respectively.
61. If a, b, c, d and p are different real numbers such that $(a^2 + b^2 + c^2)p^2 - 2(ab + bc + cd)p + (b^2 + c^2 + d^2) = 0$, show that a, b, c and d are in GP.
62. If a, b, c are respectively the p th, q th and r th terms of a GP, show that

$$(q-r) \log a + (r-p) \log b + (p-q) \log c = 0$$
63. If $(1-k)(1+2x+4x^2+8x^3+16x^4+32x^5) = 1-k^6$, where $k \neq 1$, find the value of $\frac{k}{x}$.
64. If α and β be the roots of $x^2 - 3x + a = 0$ and γ and δ be the roots of $x^2 - 12x + b = 0$ and numbers $\alpha, \beta, \gamma, \delta$ (in order) form an increasing GP, prove that the value of $a = 2$ and $b = 32$.
65. If three distinct real numbers x, y, z are in GP such that $x + y + z = ax$, find the value of a .

SUM OF n TERMS OF GP

66. Find the sum of $1 + 3 + 9 + 27 + \dots$ to n terms.
67. Find the sum of $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$ to n terms.
68. How many terms of the series $1 + 3 + 3^2 + 3^3 + \dots$ must be taken to make 3280?
69. Find the sum of the geometric series
 $(x+y) + (x^2+xy+y^2) + (x^3+x^2y+xy^2+y^3) + \dots$ to n -terms.
70. Find the sum of the geometric series

$$\frac{a}{(1+i)} + \frac{a}{(1+i)^2} + \frac{a}{(1+i)^3} + \dots + \frac{a}{(1+i)^n}.$$

71. Find the value of $\sum_{k=1}^{10} (2+3^k)$.

72. Evaluate $\sum_{n=1}^n (2^{n-1} + 3^n)$.

73. Evaluate

(i) $\sum_{k=1}^n (4^{k-1} + 5^{k+1})$

(ii) $\sum_{n=1}^n \left[\left(\frac{1}{3} \right)^{n-1} + \left(\frac{1}{5} \right)^{n+1} \right]$

74. Find the sum of the following series

(i) $5 + 55 + 555 + \dots$ to n terms

(ii) $7 + 77 + 777 + \dots$ to n terms

(iii) $9 + 99 + 999 + \dots$ to n terms

75. Find the sum of $(6666 \dots 6)^2 + (8888 \dots 8)^2$ (upto n digits).

76. Find the sum $\sum_{n=1}^{10} \left[\left(\frac{1}{2} \right)^{n-1} + \left(\frac{1}{5} \right)^{n+1} \right]$.

77. Prove that the sum to n terms of the series $11 + 103 + 1005 + \dots$ is $\frac{10}{9} (10^n - 1) + n^2$.

78. Find the least value of n for which the sum $1 + 3 + 3^2 + \dots$ to n terms is greater than 7000.

79. Find the sum of n terms of the series
 $\frac{1}{3} + \frac{5}{9} + \frac{19}{27} + \frac{65}{81} + \dots$ to n terms.

80. If $S = \frac{2}{3} + \frac{8}{9} + \frac{26}{27} + \frac{30}{81} + \dots$ to n terms, find the value of S .

81. If S be the sum, P the product and R the sum of the reciprocals of n terms of a GP, prove that $\left(\frac{S}{R} \right)^n = P^2$.

82. If f is a function satisfying $f(x+y) = f(x) + f(y)$ for all $x, y \in N$ such that $f(1) = 3$ and $\sum_{x=1}^n f(x) = 120$, find the value of n .

83. Let a_n be the n th term of the GP of positive numbers. Let $\sum_{n=1}^{100} a_{2n} = \alpha$ and $\sum_{n=1}^{100} a_{2n-1} = \beta$ such that $\alpha \neq \beta$.

Prove that the common ratio of the GP is $\frac{\alpha}{\beta}$.

SUM OF AN INFINITE GP

84. Find the sum of

$$(\sqrt{2}+1) + 1 + (\sqrt{2}-1) + \dots \text{to } \infty$$

85. Find the sum of

$$\frac{1}{2} + \frac{1}{3^2} + \frac{1}{2^3} + \frac{1}{3^4} + \frac{1}{2^5} + \frac{1}{3^6} + \dots \infty$$

86. If $b = a + a^2 + a^3 + \dots \infty$, prove that $a = \frac{b}{1+b}$.

87. If $x = a + \frac{a}{r} + \frac{a}{r^2} + \dots \infty$, $y = b - \frac{b}{r} + \frac{b}{r^2} - \dots \infty$ and,

$$z = c + \frac{c}{r^2} + \frac{c}{r^4} + \dots \text{ prove that } \frac{xy}{z} = \frac{ab}{c}.$$

88. If $x = 1 + a + a^2 + \dots \infty$, where $|a| < 1$ and $y = 1 + b + b^2 + \dots \infty$, where $|b| < 1$. Prove that

$$1 + ab + (ab)^2 + \dots \infty = \frac{xy}{x + y - 1}$$

89. If $A = 1 + r^a + r^{2a} + \dots$ to ∞ and $B = 1 + r^b + r^{2b} + \dots \infty$, prove that

$$r = \left(\frac{A-1}{A} \right)^{1/a} = \left(\frac{B-1}{B} \right)^{1/b}$$

90. If $x = \sum_{n=0}^{\infty} \cos^{2n} \theta$, $y = \sum_{n=0}^{\infty} \sin^{2n} \theta$, $z = \sum_{n=0}^{\infty} \cos^{2n} \theta \sin^{2n} \theta$,

where $0 < \theta, \varphi < \frac{\pi}{2}$, prove that $xz + yz - z = xy$.

91. If $|x| < 1$ and $|y| < 1$, find the sum

$$(x+y) + (x^2+xy+y^2) + (x^3+x^2y+xy^2+y^3) + \dots \text{ to } \infty$$

92. If S_p denotes the sum of the series $1 + r^p + r^{2p} + \dots$ to ∞ and s_p the sum of the series $1 - r^p + r^{2p} - \dots$ to ∞ prove that $S_p + s_p = 2 \cdot S_{2p}$.

93. Let $S \subset (-\pi, \pi)$ denotes the set of values of x satisfying the equation $8^{1+|\cos x|+|\cos x|^2+|\cos x|^3+\dots \text{ to } \infty} = 4^3$, prove

$$\text{that } S = \left(-\frac{\pi}{3}, -\frac{2\pi}{3} \right).$$

94. If $\exp\{(\sin^2 x + \sin^4 x + \sin^6 x + \dots \infty)\}$ satisfies the equation $x^2 - 9x + 8 = 0$, prove that the value of

$$\left(\frac{\cos x}{\cos x + \sin x} \right) = \frac{1}{2}(\sqrt{3} - 1).$$

95. Find the value of $(0.2)^{\log_{\sqrt{2}} \left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \text{inf} \right)}$.

PROPERTIES OF GP

96. If p, q, r are in an AP, show that the p th, q th and r th terms of a GP are in GP.

97. If a, b, c are in AP and x, y, z are in G.P., show that $x^{b-c} \cdot y^{c-a} \cdot z^{a-b} = 1$.

98. If a, b, c are in GP, prove that $\log(a^n), \log(b^n), \log(c^n)$ are in AP.

99. If a, b, c are in GP and x, y are the arithmetic means of a, b and b, c respectively, prove that $\frac{a}{x} + \frac{c}{y} = 2$ and

$$\frac{1}{x} + \frac{1}{y} = \frac{2}{b}.$$

100. If $a, b, 3c$ are in AP and $a, b, 4c$ are in GP, find the value of a/b .

101. If $x^a = x^{\frac{b}{2}} z^{\frac{b}{2}} = z^c$, prove that $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in AP.

102. If a, b, c are three distinct real numbers in GP and $a + b + c = xb$, prove that $x < -1$ or $x > 3$.

103. If a, b, c in GP, prove that

(i) $a(b^2 + c^2) = c(a^2 + b^2)$

(ii) $a^2 b^2 c^2 \left(\frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3} \right) = a^3 + b^3 + c^3$

(iii) $\frac{(a+b+c)^2}{a^2+b^2+c^2} = \frac{a+b+c}{a-b+c}$

(iv) $\frac{1}{a^2-b^2} + \frac{1}{b^2} = \frac{1}{b^2-c^2}$

(v) $(a+2b+2c)(a-2b+2c) = a^2 + 4c^2$

104. Suppose a, b, c are in AP and a^2, b^2, c^2 are in GP. If $a < b < c$ and $a + b + c = \frac{3}{2}$, find the value of a .

INSERTION OF GM

105. Find the value of n so that $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ may be the geometric mean between a and b .

106. Find two positive numbers whose difference is 12 and whose AM exceeds the GM by 2.

107. Let x be the arithmetic mean, and y and z be two geometric means between any two positive numbers.

Prove that $\frac{y^3 + z^3}{xyz} = 2$.

108. If a be the AM of b and c and the two geometric means are G_1 and G_2 , prove that $G_1^3 + G_2^3 = 2abc$.

109. If one geometric mean G and two arithmetic means A_1 and A_2 be inserted between two given quantities, prove that $G^2 = (2A_1 - A_2)(2A_2 - A_1)$.

110. If a, b, c are in GP and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, show that $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in AP.

111. The sum of two numbers is 6 times their geometric means, show that the numbers are in the ratio $(3 + 2\sqrt{2}) : (3 - 2\sqrt{2})$.

HARMONIC PROGRESSION

112. If m th term of an HP is n and n th term is m , prove that (mn) th term is 1.

113. If m th term of an HP is n and n th term is m , prove that $(m+n)$ th term is $\frac{mn}{m+n}$.

114. If p th term of an HP is $1/q$ and q th term is $1/p$, prove that n th term is $\frac{1}{p+q-n}$.

115. If a_1, a_2, a_3 are in AP; a_2, a_3, a_4 are in GP and a_3, a_4, a_5 are in HP, prove that a_1, a_3, a_5 are in GP.

116. If $a^x = b^y = c^z = d^w$ and a, b, c, d are in GP, prove that x, y, z, w are in HP

117. If $x > 1, y > 1$ and $z > 1$ are in GP, prove that $\frac{1}{1 + \ln x}, \frac{1}{1 + \ln y}, \frac{1}{1 + \ln z}$ are in HP.

118. If p th, q th and r th terms of an HP are a, b, c , respectively, prove that $\frac{q-r}{a} + \frac{r-p}{b} + \frac{p-q}{c} = 0$.

119. If $x_1, x_2, x_3, \dots, x_n$ are in HP, prove that $x_1 x_2 + x_2 x_3 + \dots + x_{n-1} x_n = (n-1)x_1 x_n$

120. Let a, b, c be three distinct real numbers in AP such that $|a| < 1, |b| < 1, |c| < 1$, if

$$x = \sum_{n=0}^{\infty} a^n, y = \sum_{n=0}^{\infty} b^n, z = \sum_{n=0}^{\infty} c^n,$$

then prove that x, y, z are in HP.

121. Let $a_1, a_2, a_3, \dots, a_{10}$ are in AP and $h_1, h_2, h_3, \dots, h_{10}$ are in HP. If $a_1 = 2 = h_1$ and $a_{10} = 3 = h_{10}$, find the value of $a_4 h_7 + 2007$.

122. If $a^2 + 9b^2 + 25c^2 = abc \left(\frac{15}{a} + \frac{5}{b} + \frac{3}{c} \right)$, prove that a, b, c are in HP

123. If a, x, b are in AP, a, y, b are in GP and a, z, b are in HP such that $x = 9z$, where $a, b > 0$, prove that $y = 3|z|$ and $x = 3|y|$.

124. If $A_1 A_2; G_1 G_2; H_1 H_2$ are arithmetic, geometric and harmonic means, respectively between two quantities a and b , prove that

$$(i) \frac{G_1 G_2}{H_1 H_2} = \frac{A_1 + A_2}{H_1 + H_2}$$

$$(ii) A_1 H_2 = A_2 H_1 = G_1 G_2 = ab.$$

125. If $4a^2 + 9b^2 + 16c^2 = 2(3ab + 6bc + 4ac)$, where a, b, c are non-zero real numbers, prove that a, b, c are in HP.

126. If a, b, c are in HP such that $\left(\frac{a+b}{2a-b} \right) + \left(\frac{c+b}{2c-b} \right) > \lambda$, find the value of λ .

127. If a, b, c are three positive real numbers, prove that $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 3$.

128. If a, b, c are in HP, find the value of $\left(\frac{1}{b} + \frac{1}{c} - \frac{1}{a} \right) \left(\frac{1}{c} + \frac{1}{a} - \frac{1}{b} \right)$.

129. If a, b, c, d are in HP, prove that $ab + bc + cd = 3ad$.

130. If a, b, c are in AP; x, y, z are in HP; ax, by, cz are in GP, prove that $\frac{x}{z} + \frac{z}{x} = \frac{a}{c} + \frac{c}{a}$.

131. If $\frac{a-x}{px} = \frac{a-y}{qy} = \frac{a-z}{rz}$ and p, q, r are in AP, prove that x, y, z are in HP.

ARITHMETICO GEOMETRIC PROGRESSION

132. Find the n th term of the following arithmetico geometric series.

$$(i) 1 + 5x + 9x^2 + 13x^3 + \dots$$

$$(ii) 1 - 3x + 5x^2 - 7x^3 + \dots$$

$$(iii) 1 + \frac{2}{3} + \frac{3}{3^2} + \frac{4}{3^3} + \dots$$

133. Find the sum of n terms of the following series.

$$(i) 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$(ii) 1 + 4x + 7x^2 + 10x^3 + \dots$$

$$(iii) 1 + \frac{3}{2} + \frac{5}{4} + \frac{7}{8} + \dots$$

134. Find the sum to infinity of the following series when $|x| < 1$.

$$(i) 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$(ii) 1 + 3x + 5x^2 + 7x^3 + \dots$$

$$(iii) 1 + 4x + 7x^2 + 10x^3 + \dots$$

135. Find the sum to n terms of the series

$$(i) 1 + \frac{3}{2} + \frac{5}{4} + \frac{7}{8} + \frac{9}{16} + \dots$$

$$(ii) \frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{16} + \dots$$

$$(iii) \frac{2}{3} + \frac{8}{9} + \frac{26}{27} + \frac{80}{81} + \dots$$

$$(iv) 2 + \frac{7}{4} + \frac{10}{8} + \frac{13}{16} + \frac{16}{32} + \dots$$

136. If the sum to infinity of the series

$$(i) 3 + (3+d)\frac{1}{4} + (3+2d)\frac{1}{4^2} + \dots \text{ to } \infty \text{ is } \frac{44}{9},$$

find d .

$$(ii) 1 + 2r + 3r^2 + 4r^3 + \dots \text{ is } 9/4, \text{ find } r.$$

SOME SPECIAL SEQUENCES

137. Find the sum to n terms of

$$1.2 + 2.3 + 3.4 + 5.6 + \dots$$

138. Find the sum to n terms of

$$1.4 + 3.7 + 5.10 + 7.13 + \dots$$

139. Find the sum to n terms of

$$1.2.3 + 2.3.4 + 3.4.5 + \dots$$

140. Find the sum to n terms of

$$1.3.5 + 2.5.7 + 3.7.9 + \dots$$

141. Find the sum to n terms of

$$1 + (1+2) + (1+2+3) + (1+2+3+4) + \dots$$

142. Find the sum to n terms of

$$1 + (1+3) + (1+3+5) + (1+3+5+7) + \dots$$

143. Find the sum to n terms of

$$1^2 + (1^2 + 2^2) + (1^2 + 2^2 + 3^2) + (1^2 + 2^2 + 3^2 + 4^2) + \dots$$

144. Find the sum to n terms of

$$\frac{1^2}{1} + \left(\frac{1^2 + 2^2}{1+2} \right) + \left(\frac{1^2 + 2^2 + 3^2}{1+2+3} \right) + \dots$$

145. Find the sum to
- n
- terms of

$$\frac{1^2}{1} + \left(\frac{1^3 + 2^3}{1+2} \right) + \left(\frac{1^3 + 2^3 + 3^3}{1+2+3} \right) + \dots$$

146. Find the sum to
- n
- terms of

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \frac{1}{4.5} + \dots$$

147. Find the sum to
- n
- terms of

$$\frac{1}{1.3} + \frac{1}{4.5} + \frac{1}{7.7} + \frac{1}{10.9} + \dots$$

148. Find the sum to
- n
- terms of

$$\frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \dots$$

149. Find the sum to
- n
- terms of

$$\frac{1}{1.3.5} + \frac{1}{3.5.7} + \frac{1}{5.7.9} + \dots$$

150. Find the sum to
- n
- terms of

$$\frac{1}{1.4.7} + \frac{1}{4.7.10} + \frac{1}{7.10.13} + \dots$$

151. Prove that
- $\frac{1 \times 2^2 + 2 \times 3^2 + \dots + n \times (n+1)^2}{1^2 \times 2 + 2^2 \times 3 + \dots + n^2 \times (n+1)} = \frac{3n+5}{3n+1}$
- .

152. Find the sum to
- n
- terms of the series

$$\frac{4}{2.3.4} + \frac{7}{3.4.5} + \frac{10}{4.5.6} + \dots$$

153. Find the sum of the series

$$\frac{1}{\log_2 4} + \frac{1}{\log_{2^2} 4} + \frac{1}{\log_{2^3} 4} + \dots + \frac{1}{\log_{2^n} 4}$$

154. Find the value of the expression
- $\sum_{i=1}^n \sum_{j=1}^i \sum_{k=1}^j 1$
- .

155. Find the sum of
- n
- terms of the series

$$3 + 7 + 13 + 21 + \dots$$

156. Prove that the sum of the series

$$1 + 3 + 7 + 15 + 31 + \dots \text{ to } n \text{ terms} = 2^{n+1} - n = 2.$$

157. Find the sum to
- n
- terms of the series

$$1 + 4 + 10 + 22 + \dots$$

A.M/G.M/H.M

158. If
- $x, y, z > 0$
- and
- $x + y + z = 1$
- , find the maximum value of
- $\frac{xyz}{(1-x)(1-y)(1-z)}$
- .

159. If
- p
- and
- q
- are positive real numbers such that
- $p + q = 1$
- , prove that

$$\left(p + \frac{1}{p} \right)^2 + \left(q + \frac{1}{q} \right)^2 \geq \frac{25}{2}.$$

160. If
- a, b, c
- are positive real numbers such that
- $a + b + c = 1$
- , prove that

$$\left(1 + \frac{1}{a} \right) \left(1 + \frac{1}{b} \right) \left(1 + \frac{1}{c} \right) \geq 64.$$

161. If
- a, b, c, d
- are positive real numbers, prove that
- $(a+b+c+d) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right) \geq \frac{1}{16}$
- .

162. If
- a, b, c
- are positive real numbers, prove that
- $(a+b+c)(ab+bc+ca) \geq 9abc$
- .

163. If
- $(1+x)(1+x^2)(1+x^4) \dots (1+x^{128}) = \sum_{r=0}^n x^r$
- , find the value of
- n
- .

164. Let
- x
- be arithmetic mean, and
- y
- , and
- z
- be two geometric means between any two positive numbers, find the value of
- $\frac{y^3 + z^3}{xyz}$
- .

165. The sides of a right-angled triangle form a GP. The shortest side has length 2. The length of the hypotenuse is of the form
- $a + \sqrt{b}$
- , where
- $a, b \in \mathbb{N}$
- , find the value of
- $a^2 + b^2 + 10$
- .

166. If
- a, b, c
- are the sides of a triangle, prove that
- $\frac{1}{2} < \frac{ab+bc+ca}{a^2+b^2+c^2} < 1$
- .

167. If
- a, b, c, d
- are four positive real numbers such that
- $abcd = 1$
- , prove that
- $(1+a)(1+b)(1+c)(1+d) \geq 16$
- .

168. If
- $a > 1, b > 1, c > 1$
- and
- $d > 1$
- , prove that
- $(1+a)(1+b)(1+c)(1+d) < 8(abcd + 1)$
- .

169. If
- $x + y + z = 1$
- , prove that

$$(1-x)(1-y)(1-z) < \frac{8}{27}.$$

170. If
- a, b, c
- are positive real numbers such that
- $a + b + c = 1$
- , prove that

$$\frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} \geq 27$$

171. If
- $a^2 + b^2 = 1$
- and
- $x^2 + y^2 = 1$
- , prove that
- $(ax + by) < 1$
- .

172. If
- p, q, x, y
- are positive, prove that
- $(px + qy)(pq + xy) \geq 4pqxy$
- .

173. If
- $\log_2 x + \log_2 y \geq 6$
- , prove that the least value of
- $x + y$
- is 16.

174. If
- $a > 1, b > 1$
- , prove that
- $\log_b a + \log_a b \geq 2$
- .

175. If
- $x > 1$
- , find the least value of
- $2\log_{10} x - \log_x (.01)$
- .

176. If
- $a^2 + b^2 = 1$
- , prove that

$$\left(a^2 - \frac{1}{a^2} \right)^2 + \left(b^2 - \frac{1}{b^2} \right)^2 \geq \frac{9}{2}.$$

177. Let
- $a, b, c, d > 0$
- and
- $a \neq b \neq c \neq d$
- such that
- $m = a + b + c + d$
- , prove that
- $(m-a)(m-b)(m-c)(m-d) > 81abcd$
- .

178. If
- $a + b = 1$
- , find the least value of
- $\left(a + \frac{1}{a} \right)^2 + \left(b + \frac{1}{b} \right)^2$
- .

179. If
- $x^2 + y^2 = 2$
- and
- $a^2 + b^2 = 8$
- , find the greatest value of
- $ax + by$
- .

180. If $a > 0$, $b > 0$, $c > 0$ and $a + b + c = 1$, find the least value of

$$\left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 + \left(c + \frac{1}{c}\right)^2.$$

181. If $a + 2b + 3c = 12$ and $a > 0$, $b > 0$, $c > 0$, find the greatest value of ab^2c^3 .
 182. If $4a + 3b + 2c = 45$ and $a > 0$, $b > 0$, $c > 0$, find the greatest value of $a^2b^3c^4$.
 183. If $2x + 3y = 10$, find the maximum value of x^2y^3 .
 184. If $x + 5y = 18$, find the greatest value of xy^5 .
 185. If three positive real numbers a, b, c are in AP with $abc = 64$, find the minimum value of b .
 186. Find the minimum value of

$$f(x) = x^2 + 2 + \frac{1}{x^2 + 1}, x > 0.$$

187. Find the minimum value of

$$f(x) = x^{10} + \frac{10}{x}, x > 0.$$

188. Find the minimum value of

$$f(x) = x^{2012} + \frac{2012}{x}, x > 0$$

189. Find the minimum value of

$$f(a) = a^9 + a^7 + a^5 + a^3 + 1 + a^{-3} + a^{-5} + a^{-7} + a^{-9}$$

where $a > 0$.

190. Find the minimum value of

$$f(x) = 2^{x^2+2x+2} + 2^{2-2x-x^2}, x > 0.$$

191. Find the minimum value of

$$f(x) = 2^x + 3^x + 4^x + 5^x + 2^{-x} + 3^{-x} + 4^{-x} + 5^{-x} + 10$$

192. Let a, b, c, d are all positive real numbers such that $a + 2b + 3c + 4d = 10$. If M is the maximum value of $(a + 2b)(3c + 4d)$ and N is the maximum value of $(a + c + 2d)(b + c + d)$ then find the value of $(M + 2N + 10)$.

193. Let $a + b + c = 1$ such that $a > 0$, $b > 0$, $c > 0$, find the minimum value of $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$.

194. Let $x + y + z = 1$ such that $x > 0$, $y > 0$, $z > 0$, find the least value of $(x^3 + y^3 + z^3)(x^6 + y^6 + z^6)$.

195. If $a > 0$, $b > 0$, $c > 0$ and $abc = 8$, find the minimum value of $(1 + a + a^2)(1 + b + b^2)(1 + c + c^2)$.

196. Let $x > 0$, $y > 0$, $z > 0$ with $xyz = 2$, find the minimum value of $(1 + x^3)(1 + y^3)(1 + z^3)$.

197. If a, b are positive real numbers, prove that

$$\{(1+a)(1+b)\}^3 \geq \frac{3^6}{16}(a^2b^2).$$

198. If a, b, c are positive real numbers, prove that

$$(1+a)(1+b)(1+c)^7 > 7^7 a^4 b^4 c^4.$$

199. If a, b, c are positive real numbers such that $a + b + c = 1$,

$$\text{find the minimum value of } \frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca}.$$

200. If a, b, c, d are positive real numbers such that $a + b + c + d + e = 8$ and $a^2 + b^2 + c^2 + d^2 + e^2 = 16$, find the range of e .

201. Let a, b, c are positive real numbers with $ab + bc + ca = 8$. Find the maximum value of abc .

202. Let $\alpha, \beta, \gamma, \delta$ are four positive roots of $x^4 + bx^3 + cx^2 + dx + 6 = 0$, find the minimum value of (bd) .

203. If a, b, c are real numbers such that $a + 2b + c = 4$, find the maximum value of $ab + bc + ca$.

204. If a, b, c, d are positive real numbers such that $a + b + c + d = 2$, find the maximum value $(a + b)(c + d)$.

205. If $a + b + c = 1$, find the minimum value of $\left(\frac{1}{a} - 1\right)\left(\frac{1}{b} - 1\right)\left(\frac{1}{c} - 1\right)$.

206. If a, b, c are all positive, prove that

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

207. If x, y, z, w are all greater than 1, prove that

$$(1+x)(1+y)(1+z)(1+w) < 4(xy+1)(zw+1).$$

208. If x, y, z and w are all greater than 1, prove that $(1+x)(1+y)(1+z)(1+w) < 8(xyzw+1)$.

209. If a, b, c are any three real numbers, show that $a^4 + b^4 + c^2 \geq 2\sqrt{2}abc$.

210. If a, b, c are positive real numbers, prove that

$$\frac{ab}{c} + \frac{bc}{a} + \frac{ca}{b} \geq a + b + c$$

211. If a, b, c are positive real numbers, prove that

$$\left(\frac{b^2+c^2}{b+c} + \frac{c^2+a^2}{c+a} + \frac{a^2+b^2}{a+b}\right) \geq (a+b+c).$$

212. Find the least value of $\left(\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2}\right)$ where $x^2 + y^2 + z^2 = 1$.

213. Find the minimum value of the sum of real numbers $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ and a^{10} with $a > 0$.

Solutions:

4. Let the first term = a and the common difference = d

$$t_m = \frac{1}{n} \Rightarrow a + (m-1)d = \frac{1}{n}$$

$$t_n = \frac{1}{m} \Rightarrow a + (n-1)d = \frac{1}{m}$$

On subtracting, we get

$$(m-n)d = \frac{1}{n} - \frac{1}{m} = \frac{m-n}{mn}$$

$$\Rightarrow d = \frac{1}{mn}$$

$$\text{Also, } a + (m-1)\frac{1}{mn} = \frac{1}{n}$$

$$\Rightarrow a = \frac{1}{mn}$$

$$\text{Thus, } t_{mn} = a + (mn-1)d$$

$$= \frac{1}{mn} + (mn-1)\frac{1}{mn}$$

$$= \frac{1 + (mn-1)}{mn}$$

$$= 1$$

5. Let the first term = a and the common difference = d

$$\text{We have } t_{m-1} + 2t_{n+1}$$

$$\Rightarrow a + md = 2(a + nd)$$

$$\Rightarrow a = (m-2n)d$$

$$\text{Now, } t_{3m+1} = a + 3md$$

$$= (m-2n)d + 3md = 2(2m-n)d$$

$$\text{and } t_{m+n+1} = a + (m+n)d$$

$$= (m-2n)d + (m+n)d$$

$$= (2m-n)d$$

Hence, the result.

6. Let the first term = a and the common difference = d .

$$\text{We have } T_p = q \Rightarrow a + (p-1)d = q$$

$$\text{and } T_{p+q} = 0 \Rightarrow a + (p+q-1)d = 0$$

On subtracting, we get

$$\{(p-1) - (p+q-1)\}d = q$$

$$\Rightarrow d = -1$$

Put $d = -1$ in eq.(i), we get

$$a - (p-1) = q$$

$$\Rightarrow a = p + q - 1$$

$$\text{Now, } T_q = a + (q-1)d$$

$$= p + q - 1 - (q-1)$$

$$= p$$

7. Do yourself.

8. Let the first term = a and the common difference = d

We have

$$t_{m-n} + t_{m+n} = a + (m-n-1)d + a + (m+n-1)d$$

$$= 2a + (m-n-1 + m+n-1)d$$

$$= 2a + (2m-1)d$$

$$= 2(a + (m-1)d)$$

$$= 2t_m$$

9. (i) Let the first term = A and the common difference = D

It is given that,

$$t_p = A + (p-1)D = a$$

$$t_q = A + (q-1)D = b$$

$$t_r = A + (r-1)D = c$$

$$\text{Now, } a(q-r) + b(r-p) + c(p-q)$$

$$= \{A + (p-1)D\}(q-r)$$

$$+ \{A + (q-1)D\}(r-p)$$

$$+ \{A + (r-1)D\}(p-q)$$

$$= A(q-r+r-p+p-q)$$

$$- D(q-r+r-p+p-q)$$

$$+ \{p(q-r) + q(r-p) + r(p-q)\}$$

$$= 0 + 0 + 0$$

$$= 0.$$

10. We have

$$\frac{a_4}{a_7} = \frac{2}{3}$$

$$\Rightarrow \frac{a+3d}{a+6d} = \frac{2}{3}$$

$$\Rightarrow 3a+9d = 2a+12d$$

$$\Rightarrow a = 3d$$

$$\text{Now, } \frac{a_6}{a_8} = \frac{a+5d}{a+7d}$$

$$= \frac{3d+5d}{3d+7d}$$

$$= \frac{8d}{10d}$$

$$= \frac{4}{5}$$

11. Since a, b, c, d and e are in AP, so

$$a + e = b + d = 2c$$

We have

$$a - 4b + 6c - 4d + e = (a + e) - 4(b + d) + 6c$$

$$= 2c - 4(2c) + 6c$$

$$= 8c - 8c$$

$$= 0$$

12. Do yourself

13. Let $ABCD$ be a quadrilateral in which $\angle A, \angle B, \angle C, \angle D$ are in AP.

Let $\angle A = a - 3d, \angle B = a - d, \angle C = a + d, \angle D = a + 3d$ where $2d$ is the common difference.

It is given that $2d = 10, \Rightarrow d = 5$

Clearly, $a = 90^\circ$

Thus, the angles are

$$\angle A = 90^\circ - 15^\circ = 75^\circ$$

$$\angle B = 90^\circ - 5^\circ = 85^\circ$$

$$\angle C = 90^\circ + 5^\circ = 95^\circ$$

$$\angle D = 90^\circ + 15^\circ = 105^\circ$$

Hence, the angles are

$$75^\circ, 85^\circ, 95^\circ \text{ and } 105^\circ$$

14. Let the roots be $a-d, a, a+d$

$$\text{So, } a-d + a + a+d = 12$$

$$\Rightarrow 3a = 12$$

$$\Rightarrow a = 4$$

$$\text{Also, } (a-d) \cdot a \cdot (a+d) = 28$$

$$\begin{aligned}
&\Rightarrow a(a^2 - d^2) = 28 \\
&\Rightarrow 4(16 - d^2) = 28 \\
&\Rightarrow (16 - d^2) = 7 \\
&\Rightarrow d^2 = 9 \\
&\Rightarrow d = 3
\end{aligned}$$

15. Do yourself.

16. We have

$$\begin{aligned}
&\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \dots + \frac{1}{a_{n-1} a_n} \\
&= \frac{1}{d} \left(\frac{d}{a_1 a_2} + \frac{d}{a_2 a_3} + \dots + \frac{d}{a_{n-1} a_n} \right) \\
&= \frac{1}{d} \left(\frac{a_2 - a_1}{a_1 a_2} + \frac{a_3 - a_2}{a_2 a_3} + \dots + \frac{a_n - a_{n-1}}{a_{n-1} a_n} \right) \\
&= \frac{1}{d} \left(\frac{1}{a_1} - \frac{1}{a_2} + \frac{1}{a_2} - \frac{1}{a_3} + \dots + \frac{1}{a_{n-1}} - \frac{1}{a_n} \right) \\
&= \frac{1}{d} \left(\frac{1}{a_1} - \frac{1}{a_n} \right) \\
&= \frac{1}{d} \left(\frac{a_n - a_1}{a_1 a_n} \right) \\
&= \frac{1}{d} \left(\frac{\{a_1 + (n-1)d\} - a_1}{a_1 a_n} \right) \\
&= \frac{(n-1)}{a_1 a_n}
\end{aligned}$$

Hence, the result.

17. Given $a_1, a_2, \dots, a_n$ are in AP. So,

$$a_2 - a_1 = a_3 - a_2 = \dots = a_{n-1} - a_n = d(\text{say})$$

We have

$$\begin{aligned}
&\frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}} \\
&= \left(\frac{\sqrt{a_2} - \sqrt{a_1}}{a_2 - a_1} + \frac{\sqrt{a_3} - \sqrt{a_2}}{a_3 - a_2} + \dots + \frac{\sqrt{a_n} - \sqrt{a_{n-1}}}{a_n - a_{n-1}} \right) \\
&= \left(\frac{\sqrt{a_2} - \sqrt{a_1}}{d} + \frac{\sqrt{a_3} - \sqrt{a_2}}{d} + \dots + \frac{\sqrt{a_n} - \sqrt{a_{n-1}}}{d} \right) \\
&= \frac{1}{d} (\sqrt{a_n} - \sqrt{a_1}) \\
&= \frac{1}{d} \left(\frac{a_n - a_1}{\sqrt{a_n} + \sqrt{a_1}} \right) \\
&= \frac{1}{d} \left(\frac{a_1 + (n-1)d - a_1}{\sqrt{a_n} + \sqrt{a_1}} \right) \\
&= \left(\frac{(n-1)}{\sqrt{a_n} + \sqrt{a_1}} \right)
\end{aligned}$$

18. Given $a_1, a_2, \dots, a_n$ are in AP. So,

$$a_2 - a_1 = a_3 - a_2 = \dots = a_{n-1} - a_n = d(\text{say})$$

We have

$$\begin{aligned}
&\frac{1}{a_1 a_n} + \frac{1}{a_2 a_{n-1}} + \frac{1}{a_3 a_{n-2}} + \dots + \frac{1}{a_n a_1} \\
&= \frac{1}{a_1 + a_n} \left(\frac{a_1 + a_n}{a_1 a_n} + \frac{a_1 + a_n}{a_2 a_{n-1}} + \frac{a_1 + a_n}{a_3 a_{n-2}} \right. \\
&\quad \left. + \dots + \frac{a_1 + a_n}{a_n a_1} \right) \\
&= \frac{1}{a_1 + a_n} \left(\frac{a_1 + a_n}{a_1 a_n} + \frac{a_2 + a_{n-1}}{a_2 a_{n-1}} + \frac{a_3 + a_{n-2}}{a_3 a_{n-2}} \right. \\
&\quad \left. + \dots + \frac{a_1 + a_n}{a_n a_1} \right) \\
&= \frac{1}{a_1 + a_n} \left(\left(\frac{1}{a_1} + \frac{1}{a_n} \right) + \left(\frac{1}{a_2} + \frac{1}{a_{n-1}} \right) \right. \\
&\quad \left. + \left(\frac{1}{a_3} + \frac{1}{a_{n-2}} \right) + \left(\frac{1}{a_1} + \frac{1}{a_n} \right) \right) \\
&= \frac{2}{a_1 + a_n} \left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n} \right)
\end{aligned}$$

Hence, the result.

19. Let the first term be a and the common difference be d .

It is given that,

$$t_7 = 15 \Rightarrow a + 6d = 15$$

Let $P = t_2 t_7 t_{12}$

$$= (a + d)(a + 6d)(a + 11d)$$

$$= 15(15 - 5d)(15 + 5d)$$

$$= 15(225 - 25d^2)$$

$$\Rightarrow \frac{dP}{dd} = -30 \times 25d$$

$$\Rightarrow \frac{d^2 P}{dd^2} = -30 \times 25 < 0$$

So, the product is maximum.

$$\therefore d = 0.$$

20. (i) Given a, b and c are in AP.

$$\therefore 2b = a + c$$

Also, $\tan^{-1} a, \tan^{-1} b$ and $\tan^{-1} c$ are in AP.

$$\Rightarrow 2 \tan^{-1} b = \tan^{-1} a + \tan^{-1} c$$

$$\Rightarrow \tan^{-1} \left(\frac{2b}{1-b^2} \right) = \tan^{-1} \left(\frac{a+c}{1-ac} \right)$$

$$\Rightarrow \left(\frac{2b}{1-b^2} \right) = \left(\frac{a+c}{1-ac} \right)$$

$$\Rightarrow \left(\frac{2b}{1-b^2} \right) = \left(\frac{2b}{1-ac} \right)$$

$$\Rightarrow 1 - b^2 = 1 - ac$$

$$\Rightarrow b^2 = ac$$

$\Rightarrow a, b$ and c are in GP.

(ii) Also, $b^2 = ac$

$$\Rightarrow \left(\frac{a+c}{2}\right)^2 = ac$$

$$\Rightarrow (a+c)^2 = 4ac$$

$$\Rightarrow (a+c)^2 - 4ac = 0$$

$$\Rightarrow (a-c)^2 = 0$$

$$\Rightarrow (a-c) = 0$$

$$\Rightarrow a = c$$

$$\text{Again, } 2b = a + c = a + a = 2a$$

$$\Rightarrow b = a$$

$$\text{Thus, } a = b = c$$

21. Do yourself

22. Do yourself

23. Do yourself

24. Given $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$

$$\Rightarrow (a_1 + a_{24}) + (a_5 + a_{20}) + (a_{10} + a_{15}) = 225$$

$$\Rightarrow (a_1 + a_{24}) + (a_1 + a_{24}) + (a_1 + a_{24}) = 225$$

$$\Rightarrow 3(a_1 + a_{24}) = 225$$

$$\Rightarrow (a_1 + a_{24}) = 75$$

$$\text{Thus, } s_{24} = \frac{24}{2} \times (a_1 + a_{24})$$

$$= 12 \times 75 = 900$$

25. (i) Given $S_n = 3n^2 + 4n$

$$\Rightarrow S_{n+1} = 3(n+1)^2 + 4(n+1)$$

$$\begin{aligned} \text{Thus, } t_n &= S_{n+1} - S_n \\ &= \{3(n+1)^2 + 4(n+1)\} - (3n^2 + 4n) \\ &= 6n + 7 \end{aligned}$$

$$\text{Therefore, } t_{10} = 60 + 7 = 67$$

(ii) Do yourself.

26. Let the first term = a and the common difference = d .

$$\text{Clearly, } a = \frac{1}{mn} = d$$

$$\text{Now, } S_{mn} = \frac{mn}{2} [2a + (mn-1)d]$$

$$= \frac{mn}{2} \left(\frac{2}{mn} + \frac{(mn-1)}{mn} \right)$$

$$= \frac{mn}{2} \left(\frac{2 + (mn-1)}{mn} \right)$$

$$= \frac{(mn+1)}{2}$$

27. Let the n th term, $t_n = x$.

$$\begin{aligned} \text{So, } x &= a + (n-1)d \\ &= 1 + (n-1)5 \\ &= 5n - 4 \end{aligned}$$

$$\text{Thus, } \frac{n}{2}(1 + 5n - 4) = 148$$

$$\Rightarrow \frac{n}{2}(5n - 3) = 148$$

$$\Rightarrow n(5n - 3) = 296$$

$$\Rightarrow 5n^2 - 3n - 296 = 0$$

$$\Rightarrow 5n^2 - 40n + 37n - 296 = 0$$

$$\Rightarrow 5n(n-8) + 37(n-8) = 0$$

$$\Rightarrow (5n+37)(n-8) = 0$$

$$\Rightarrow n = 8, -\frac{37}{5}$$

Since n is a natural number, so $n = 8$

Hence, the value of x is

$$= 5n - 4$$

$$= 5 \times 8 - 4$$

$$= 36$$

28. Do yourself.

29. Let the first term = a and the common difference = d .

$$\text{Here, } S_1 = \frac{n}{2} [2a + (n-1)d]$$

$$S_2 = \frac{2n}{2} [2a + (2n-1)d]$$

$$S_3 = \frac{3n}{2} [2a + (3n-1)d]$$

Now, $3(S_2 - S_1)$

$$= 3 \left\{ \left(\frac{2n}{2} [2a + (2n-1)d] \right) - \left(\frac{n}{2} [2a + (n-1)d] \right) \right\}$$

$$= \frac{3n}{2} \{ 2[2a + (2n-1)d] - [2a + (n-1)d] \}$$

$$= \frac{3n}{2} [2a + (3n-1)d]$$

$$= S_3$$

Hence, the result.

30. Let the first term = A and the common difference = d .

It is given that,

$$t_p = a \Rightarrow A + (p-1)d = a$$

$$t_q = b \Rightarrow A + (q-1)d = b$$

On subtraction, we get

$$(p-q)d = (a-b)$$

$$\Rightarrow d = \frac{(a-b)}{(p-q)}$$

Now,

$$S_{p+q} = \left(\frac{p+q}{2} \right) \{ 2A + (p+q-1)d \}$$

$$= \left(\frac{p+q}{2} \right) \{ A + (p-1)d + A + (q-1)d + d \}$$

$$= \left(\frac{p+q}{2} \right) \left(a + b + \left(\frac{a-b}{p-q} \right) \right)$$

Hence, the result.

31. Let the first term = a and the common difference = d .

It is given that,

$$S_m = n \Rightarrow \frac{m}{2} \{ 2a + (m-1)d \} = n$$

$$S_n = m \Rightarrow \frac{n}{2} \{ 2a + (n-1)d \} = m$$

On subtraction, we get

$$(m-n)d = \frac{2n}{m} - \frac{2m}{n}$$

$$\Rightarrow (m-n)d = \frac{2(n^2 - m^2)}{mn}$$

$$\Rightarrow d = -\frac{2(m+n)}{mn}$$

$$\text{Now, } \frac{m}{2} \left(2a - (n-1) \frac{2(m+n)}{mn} \right) = n$$

$$\Rightarrow \left(2a - (n-1) \frac{2(m+n)}{mn} \right) = \frac{2n}{m}$$

$$\Rightarrow a = \frac{n}{m} + (n-1) \frac{(m+n)}{mn}$$

$$\Rightarrow a = \frac{n^2 + mn + n^2 - m - n}{mn}$$

$$\text{Thus, } S_{(m+n)} = \left(\frac{m+n}{2} \right) [2a + (m+n-1)d]$$

$$= \left(\frac{m+n}{2} \right) \left(\frac{2(n^2 + mn + n^2 - m - n)}{mn} - \frac{2(m+n-1)(m+n)}{mn} \right)$$

$$= \left(\frac{m+n}{mn} \right) \{ (n^2 + mn + n^2 - m - n) - (m+n-1)(m+n) \}$$

$$= \left(\frac{m+n}{mn} \right) (-mn)$$

$$= -(m+n)$$

Hence, the result.

32. It is given that,

$$\frac{S_n}{S_{n'}} = \frac{7n+1}{4n+27}$$

$$= \frac{7n^2+n}{4n^2+27n}$$

$$\text{Now, } \frac{t_n}{t_{n'}} = \frac{S_n - S_{n-1}}{S_{n'} - S_{n'-1}}$$

$$= \frac{(7n^2+n) - \{7(n-1)^2 + (n-1)\}}{(4n^2+27n) - \{4(n-1)^2 + 27(n-1)\}}$$

$$= \frac{14n-6}{8n+33}$$

$$\text{Thus, } \frac{t_{11}}{t_{11'}} = \frac{14 \cdot 11 - 6}{8 \cdot 11 + 33} = \frac{154 - 6}{88 + 33} = \frac{148}{111}$$

$$33. \text{ Clearly, } \frac{t_m}{t_n} = \frac{S_m - S_{m-1}}{S_n - S_{n-1}}$$

$$= \frac{m^2 - (m-1)^2}{n^2 - (n-1)^2}$$

$$= \frac{(2m-1)}{(2n-1)}$$

Hence, the result.

34. Let the first term be a and the common difference is d and the number of sides be n .

It is given that, $a = 120^\circ$, $d = 5^\circ$

$$\text{Also, } \frac{n}{2} \{2a + (n-1)d\} = (2n-4) \times 90^\circ$$

$$\Rightarrow \frac{n}{2} \{240 + 5(n-1)\} = (2n-4) \times 90$$

$$\Rightarrow n(5n+235) = (2n-4) \times 180$$

$$\Rightarrow 5n^2 + 235n = 360n - 720$$

$$\Rightarrow 5n^2 - 125n + 720 = 0$$

$$\Rightarrow n^2 - 25n + 144 = 0$$

$$\Rightarrow (n-9)(n-16) = 0$$

$$\Rightarrow n = 9, 16$$

But $n = 16$ does not satisfy the angles property of a regular polygon.

So, $n = 9$

Hence, the number of sides is 9.

35. We have

$$S_1 = \frac{n}{2} \{2 \cdot 1 + (n-1) \cdot 1\}$$

$$S_2 = \frac{n}{2} \{2 \cdot 2 + (n-1) \cdot 3\}$$

$$S_3 = \frac{n}{2} \{2 \cdot 3 + (n-1) \cdot 5\}$$

$\vdots$

$$S_m = \frac{n}{2} \{2 \cdot m + (n-1) \cdot (2m-1)\}$$

Thus, $S_1 + S_2 + S_3 + \dots + S_m$

$$= \frac{n}{2} [2(1+2+3+\dots+m) + (n-1)\{1+3+5+\dots+(2m-1)\}]$$

$$= \frac{n}{2} \left[\frac{2m(m+1)}{2} + (n-1)m^2 \right]$$

$$= \frac{mn}{2} [(m+1) + m(n-1)]$$

$$= \frac{mn(mn+1)}{2}$$

Hence, the result.

36. Let the first term be a and the common difference be d .

It is given that

$$S_n = n^2 p \Rightarrow \frac{n}{2} (2a + (n-1)d) = n^2 p$$

$$\Rightarrow 2a + (n-1)d = 2np \quad \dots(i)$$

$$\text{Similarly, } 2a + (m-1)d = 2mp \quad \dots(ii)$$

On subtraction, we get

$$(n-m)d = 2(n-m)p$$

$$\Rightarrow d = 2p$$

put $d = 2p$ in Eq. (i), we get

$$\Rightarrow 2a + 2(n-1)p = 2np$$

$$\Rightarrow a = p$$

$$\text{Thus, } S_p = \frac{p}{2}(2a + (p-1)d)$$

$$= \frac{p}{2}(2p + 2(p-1)p)$$

$$= \frac{p}{2}(2p + 2p^2 - 2p)$$

$$= p^3$$

38. It is given that

$$\begin{aligned} S_1 &= a_1 + a_3 + a_5 + \dots + a_{2n+1} \\ &= a + (a+2d) + (a+4d) + \dots + (a+2nd) \\ &= (n+1)a + 2d(1+2+3+\dots+n) \\ &= (n+1)a + n(n+1)d \\ &= (n+1)(a+d) \end{aligned}$$

$$\begin{aligned} \text{Also, } S_2 &= a_2 + a_4 + a_6 + \dots + a_{2n} \\ &= (a+d) + (a+3d) + (a+5d) + \dots + \{a+(2n-1)d\} \\ &= na + \{1+3+5+7+\dots+(2n-1)\}d \\ &= na + n^2d \\ &= n(a+nd) \end{aligned}$$

$$\text{Hence, } \frac{S_1}{S_2} = \frac{(n+1)(a+nd)}{n(a+nd)} = \frac{(n+1)}{n}$$

$$39. \text{ We have } S_n = nP + \frac{1}{2}n(n-1)Q$$

$$= \frac{n}{2}\{2p + (n-1)Q\}$$

Clearly, the common difference = Q .

40. (i) It is given that $a, b, c \in \text{AP}$

$$\Rightarrow a - (a+b+c), b - (a+b+c), c - (a+b+c) \in \text{AP}$$

$$\Rightarrow -(b+c), -(a+c), -(a+b) \in \text{AP}$$

$$\Rightarrow (b+c), (a+c), (a+b) \in \text{AP}$$

Hence, the result.

(iii) It is given that a, b and c are in AP.

$$\Rightarrow b - a = c - b$$

$$\Rightarrow (\sqrt{b} - \sqrt{a})(\sqrt{b} + \sqrt{a}) = (\sqrt{c} - \sqrt{b})(\sqrt{c} + \sqrt{b})$$

$$\Rightarrow \frac{(\sqrt{b} - \sqrt{a})}{(\sqrt{c} + \sqrt{b})} = \frac{(\sqrt{c} - \sqrt{b})}{(\sqrt{b} + \sqrt{a})}$$

$$\Rightarrow \frac{(\sqrt{b} - \sqrt{a})}{(\sqrt{c} + \sqrt{b})(\sqrt{c} + \sqrt{a})} = \frac{(\sqrt{c} - \sqrt{b})}{(\sqrt{b} + \sqrt{a})(\sqrt{c} + \sqrt{a})}$$

$$\begin{aligned} \Rightarrow \frac{(\sqrt{b} + \sqrt{c}) - (\sqrt{c} + \sqrt{a})}{(\sqrt{c} + \sqrt{b})(\sqrt{c} + \sqrt{a})} \\ = \frac{(\sqrt{c} + \sqrt{a}) - (\sqrt{b} + \sqrt{c})}{(\sqrt{b} + \sqrt{a})(\sqrt{c} + \sqrt{a})} \end{aligned}$$

$$\Rightarrow \frac{1}{(\sqrt{c} + \sqrt{a})} - \frac{1}{(\sqrt{c} + \sqrt{b})}$$

$$= \frac{1}{(\sqrt{b} + \sqrt{a})} - \frac{1}{(\sqrt{c} + \sqrt{a})}$$

$$\Rightarrow \frac{1}{(\sqrt{b} + \sqrt{a})}, \frac{1}{(\sqrt{c} + \sqrt{a})}, \frac{1}{(\sqrt{c} + \sqrt{b})} \in \text{AP}$$

Hence, the result.

(iv) It is given that a, b and $c \in \text{AP}$

$$\Rightarrow \frac{a}{abc}, \frac{b}{abc}, \frac{c}{abc} \in \text{AP}$$

$$\Rightarrow \frac{1}{bc}, \frac{1}{ac}, \frac{1}{ab} \in \text{AP}$$

Hence, the result.

(v) It is given that a, b and $c \in \text{AP}$

$$\begin{aligned} a\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right), b\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right), \\ c\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \in \text{AP} \end{aligned}$$

$$\begin{aligned} \Rightarrow 1 + a\left(\frac{1}{b} + \frac{1}{c}\right), 1 + b\left(\frac{1}{a} + \frac{1}{c}\right), \\ 1 + c\left(\frac{1}{a} + \frac{1}{b}\right) \in \text{AP} \end{aligned}$$

$$\Rightarrow a\left(\frac{1}{b} + \frac{1}{c}\right), b\left(\frac{1}{a} + \frac{1}{c}\right), c\left(\frac{1}{a} + \frac{1}{b}\right) \in \text{AP}$$

(vi) It is given that a, b and c are in AP.

$$\begin{aligned} \text{Now, } \{(a+c)^2 - b^2\} - \{(b+c)^2 - a^2\} \\ = (a^2 - b^2) + \{(a+c)^2 - (b+c)^2\} \\ = (a^2 - b^2) + \{(a^2 - b^2) + 2c(a-b)\} \\ = 2(a^2 - b^2) + 2c(a-b) \\ = 2(a-b)(a+b+c) \end{aligned}$$

$$\begin{aligned} \text{Also, } [(a+b)^2 - c^2] - [(a+c)^2 - b^2] \\ = (b^2 - c^2) + [(a+b)^2 - (a+c)^2] \\ = 2(b^2 - c^2) + 2a(b-c) \\ = 2(b-c)(a+b+c) \\ = 2(a-b)(a+b+c) \end{aligned}$$

Hence, the result.

41. (i) It is given that a^2, b^2 and c^2 are in AP.

$$\Rightarrow b^2 - a^2 = c^2 - b^2$$

$$\Rightarrow (b+a)(b-a) = (c+b)(c-b)$$

$$\Rightarrow \frac{(b+a)}{(c-b)} = \frac{(c+b)}{(b-a)}$$

$$\Rightarrow \frac{(c-b)}{(a+b)} = \frac{(b-a)}{(b+c)}$$

$$\Rightarrow \frac{(c-b)}{(a+b)(c+a)} = \frac{(b-a)}{(b+c)(c+a)}$$

$$\Rightarrow \frac{((a+c) - (b+a))}{(a+b)(c+a)} = \frac{((b+c) - (a+c))}{(b+c)(c+a)}$$

$$\Rightarrow \frac{1}{(a+b)} - \frac{1}{(a+c)} = \frac{1}{(c+a)} - \frac{1}{(b+c)}$$

$$\Rightarrow \frac{1}{(a+b)}, \frac{1}{(a+c)}, \frac{1}{(b+c)} \in \text{AP}$$

Hence, the result.

$$(ii) \quad \frac{1}{(a+b)}, \frac{1}{(a+c)} \text{ and } \frac{1}{(b+c)} \in \text{AP}$$

$$\Rightarrow \frac{(a+b+c)}{(a+b)}, \frac{(a+b+c)}{(a+c)}, \frac{(a+b+c)}{(b+c)} \in \text{AP}$$

$$\Rightarrow 1 + \frac{c}{(a+b)}, 1 + \frac{b}{(a+c)}, 1 + \frac{c}{(b+c)} \in \text{AP}$$

$$\Rightarrow \frac{c}{(a+b)}, \frac{b}{(a+c)}, \frac{c}{(b+c)} \in \text{AP}$$

Hence, the result.

42. (i) It is given that a, b and c are in AP.

$$\begin{aligned} \text{Now, } b^2(c+a) - a^2(b+c) &= b^2c + b^2a - a^2b - a^2c \\ &= (b^2 - a^2)c + ba(b-a) \\ &= [(b+a)c + ba](b-a) \\ &= (ab + bc + ca)(b-a) \end{aligned}$$

$$\begin{aligned} \text{Also, } c^2(a+b) - b^2(c+a) &= (c^2 - b^2)a + bc(c-b) \\ &= [(c+b)a + bc](c-b) \\ &= (ab + bc + ca)(c-b) \\ &= (ab + bc + ca)(b-a) \end{aligned}$$

Hence, the result.

- (ii) It is given that a, b and c are in AP.

$$\begin{aligned} \Rightarrow 2a, 2b, 2c &\in \text{AP} \\ \Rightarrow -2a, -2b, -2c &\in \text{AP} \\ \Rightarrow (a+b+c) - 2a, (a+b+c) - 2b, & \\ & (a+b+c) - 2c \in \text{AP} \\ \Rightarrow (b+c) - a, (a+c) - b, (a+b) - c &\in \text{AP} \end{aligned}$$

- (iii) It is given that a, b and c are in AP.

$$\begin{aligned} \text{Now, } (ca - b^2) - (bc - a^2) &= c(a-b) + (a^2 - b^2) \\ &= (a+b+c)(a-b) \end{aligned}$$

$$\begin{aligned} \text{Also, } (ab - c^2) - (ac - b^2) &= a(b-c) + (b^2 - c^2) \\ &= (a+b+c)(b-c) \\ &= (a+b+c)(a-b), \end{aligned}$$

Hence $bc - a^2, ca - b^2, ab - c^2$ are also in AP.

43. Given $\frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ are in AP

$$\Rightarrow \frac{b+c-a}{a} + 2, \frac{c+a-b}{b} + 2, \frac{a+b-c}{c} + 2 \in \text{AP}$$

$$\Rightarrow \frac{b+c+a}{a}, \frac{c+a+b}{b}, \frac{a+b+c}{c} \in \text{AP}$$

$$\Rightarrow \frac{1}{a}, \frac{1}{b}, \frac{1}{c} \in \text{AP}$$

44. Do yourself

45. We have $\frac{a^{n+1} + b^{n+1}}{a^n + b^n} = \frac{a+b}{2}$

$$\Rightarrow 2a^{n+1} + 2b^{n+1} = a^{n+1} + a^n b + ab^n + b^{n+1}$$

$$\Rightarrow a^{n+1} + b^{n+1} = a^n b + ab^n$$

$$\Rightarrow a^n(a-b) = b^n(a-b)$$

$$\Rightarrow a^n = b^n$$

$$\Rightarrow \left(\frac{a}{b}\right)^n = 1 = \left(\frac{a}{b}\right)^0$$

$$\Rightarrow n = 0$$

46. Let $A_1, A_2, A_3, \dots, A_n$ be n arithmetic means are inserted between 20 and 80. So,

$$20, A_1, A_2, A_3, \dots, A_n, 80 \in \text{AP}$$

$$\text{Now, } t_{n+2} = 80$$

$$\Rightarrow 20 + (n+1)d = 80$$

$$\Rightarrow d = \frac{60}{(n+1)} \quad \dots(i)$$

Also, it is given that,

$$\frac{A_1}{A_n} = \frac{1}{3}$$

$$\Rightarrow \frac{A_n}{A_1} = 3 \quad (ii)$$

$$\Rightarrow \frac{A_n}{A_1} + 1 = 3 + 1 = 4$$

$$\Rightarrow \frac{A_n + A_1}{A_1} = 4$$

$$\Rightarrow 4A_1 = 100 \quad \text{from (ii)}$$

$$\therefore A_1 = 25$$

$$\Rightarrow 20 + d = 25$$

$$\Rightarrow d = 5 \quad \therefore a = 20$$

Putting the value of d in Eq. (i), we get

$$\frac{60}{(n+1)} = 5$$

$$\Rightarrow (n+1) = 12$$

$$\Rightarrow n = 11$$

Hence, the value of n is 11.

47. Let $A_1, A_2, A_3, \dots, A_n$ be n arithmetic means are inserted between two positive numbers, say a and b , respectively. Thus, $a, A_1, A_2, A_3, \dots, A_n, b \in \text{AP}$
Let d be the common difference.

Now,

$$\begin{aligned} A_1 + A_2 + A_3 + \dots + A_n &= \frac{n}{2}(A_1 + A_n) \\ &= \frac{n}{2}(a + d + b - d) \\ &= \frac{n}{2}(a + b) \\ &= n\left(\frac{a+b}{2}\right) \end{aligned}$$

Hence, the result.

48. Do yourself.
 49. Do yourself.
 50. Do yourself.
 51. Do yourself.

52. Let $t_n = \frac{1}{512\sqrt{2}}$

$$\Rightarrow ar^{n-1} = \frac{1}{512\sqrt{2}}$$

$$\Rightarrow \sqrt{2} \left(\frac{1}{2}\right)^{n-1} = \frac{1}{512\sqrt{2}}$$

$$\Rightarrow \left(\frac{1}{2}\right)^{n-1} = \frac{1}{512 \times 2} = \frac{1}{1024} = \frac{1}{2^{10}}$$

$$\Rightarrow \left(\frac{1}{2}\right)^{n-1} = \left(\frac{1}{2}\right)^{10}$$

$$\Rightarrow n-1 = 10$$

$$\therefore n = 11$$

Hence, the 11th term is $\frac{1}{512\sqrt{2}}$.

53. Let $t_n = \frac{1}{128}$

$$\Rightarrow ar^{n-1} = \frac{1}{128}$$

$$\Rightarrow 2 \cdot \left(\frac{1}{2}\right)^{n-1} = \frac{1}{128}$$

$$\Rightarrow \left(\frac{1}{2}\right)^{n-1} = \frac{1}{256} = \left(\frac{1}{2}\right)^8$$

$$\Rightarrow n-1 = 8$$

$$\therefore n = 9$$

Hence, the 9th term is $\frac{1}{128}$.

54. Do yourself

55. It is given that $t_3 = 3$.

$$\Rightarrow ar^2 = 3$$

Now,

$$\begin{aligned} t_1 t_2 t_3 t_4 t_5 &= a \cdot ar \cdot ar^2 \cdot ar^3 \cdot ar^4 \\ &= a^5 r^{10} \\ &= (ar^2)^5 \\ &= (3)^5 = 243 \end{aligned}$$

56. Do yourself.

57. It is given that, $a = 1$ and

$$\begin{aligned} t_5 + t_n &= 82 \\ \Rightarrow ar^4 + a &= 82 \\ \Rightarrow r^4 + 1 &= 82 \\ \Rightarrow r^4 &= 81 \\ \Rightarrow r &= \pm 3 \end{aligned}$$

58. Clearly, $t_p = a \Rightarrow AR^{p-1} = a$

$$t_q = b \Rightarrow AR^{q-1} = b$$

$$t_r = c \Rightarrow AR^{r-1} = c$$

Now, $a^{q-r} \cdot b^{r-p} \cdot c^{p-q}$

$$\begin{aligned} &= (AR^{p-1})^{q-r} \cdot (AR^{q-1})^{r-p} \cdot (AR^{r-1})^{p-q} \\ &= A^{q-r+r-p+p-q} \cdot R^{(p-1)(q-r)+(q-1)(r-p)+(r-1)(p-q)} \\ &= A^0 \cdot R^{p(q-r)+q(r-p)+r(p-q)} \cdot R^{-(q-r+r-p+p-q)} \\ &= A^0 R^0 R^0 \\ &= 1. \end{aligned}$$

59. It is given that, $t_1 = a \Rightarrow t_1 = a$, and $t_n = b \Rightarrow ar^{n-1} = b$

Now, $P = t_1 t_2 t_3 \dots t_n$

$$\begin{aligned} &= (a)(ar)(ar^2)(ar^3) \dots (ar^{n-1}) \\ &= (a)^n r^{1+2+3+\dots+(n-1)} \\ &= (a)^n r^{\frac{n-1}{2} \times (1+n-1)} \\ &= (a)^n (r^{\frac{n-1}{2}})^2 \\ &= (a)^{\frac{n}{2}} (ar^{n-1})^{\frac{n}{2}} \\ &= (a)^{\frac{n}{2}} (b)^{\frac{n}{2}} \\ &= (ab)^{\frac{n}{2}} \end{aligned}$$

$$P^2 = (ab)^n$$

Hence, the result.

60. It is given that, $t_{m+n} = p \Rightarrow ar^{m+n-1} = p$ and $t_{m-n} = q \Rightarrow ar^{m-n-1} = q$

Multiplying both, we get

$$\begin{aligned} ar^{m+n-1} \times ar^{m-n-1} &= pq \\ \Rightarrow a^2 r^{m+n-1+m-n-1} &= pq \\ \Rightarrow a^2 r^{2m-2} &= pq \\ \Rightarrow (ar^{m-1})^2 &= pq \\ \Rightarrow (ar^{m-1}) &= \sqrt{pq} \\ \therefore t_m &= \sqrt{pq} \end{aligned}$$

$$\text{Now, } \left(\frac{q}{p}\right) = \frac{ar^{m-n-1}}{ar^{m+n-1}} = r^{-2n}$$

$$\Rightarrow r = \left(\frac{p}{q}\right)^{\frac{1}{2n}}$$

Also, $ar^{m+n-1} = p$

$$\Rightarrow a \left(\frac{p}{q}\right)^{\frac{m+n-1}{2n}} = p$$

$$\Rightarrow a = q(p)^{\frac{2n}{m+n-1}-1} = qp^{\frac{n-m+1}{m+n-1}}$$

$$\begin{aligned} \text{Thus, } t_n &= ar^{n-1} = qp^{\frac{n-m+1}{m+n-1}} \times \left(\frac{p}{q}\right)^{\frac{n-1}{2n}} \\ &= p \left(\frac{q}{p}\right)^{m/2n} \end{aligned}$$

Hence, the result.

61. We have

$$\begin{aligned} & (a^2 + b^2 + c^2)p^2 - 2(ab + bc + cd)p + (b^2 + c^2 + d^2) \leq 0 \\ \Rightarrow & (a^2p^2 - 2abp + b^2) + (b^2p^2 - 2bcp + c^2) \\ & + (c^2p^2 - 2cdp + d^2) \leq 0 \\ \Rightarrow & (ap - b)^2 + (bp - c)^2 + (cp - d)^2 \leq 0 \\ \Rightarrow & (ap - b)^2 + (bp - c)^2 + (cp - d)^2 = 0 \\ \Rightarrow & (ap - b) = (bp - c) = (cp - d) = 0 \\ \Rightarrow & \frac{b}{a} = \frac{1}{p} = \frac{c}{b} = \frac{d}{c} \\ \Rightarrow & \frac{b}{a} = \frac{c}{b} = \frac{d}{c} \\ \Rightarrow & a, b, c, d \text{ are in G.P.} \end{aligned}$$

62. It is given that,

$$\begin{aligned} t_p &= AR^{p-1} = a \\ t_q &= AR^{q-1} = b \\ t_r &= AR^{r-1} = c \end{aligned}$$

$$\begin{aligned} \text{Now, } & (q-r)\log a + (r-p)\log b + (p-q)\log c \\ &= (q-r)\{\log A + (p-1)\log R\} \\ & \quad + (r-p)\{\log A + (q-1)\log R\} \\ & \quad + (p-q)\{\log A + (r-1)\log R\} \\ &= \log A \{q-r+r-p+p-q\} \\ & \quad - \log R \{q-r+r-p+p-q\} \\ & \quad + \log R \{p(q-r) + q(r-p) + r(p-q)\} \\ &= 0 + 0 + 0 \\ &= 0. \end{aligned}$$

63. We have

$$\begin{aligned} & (1-k)(1+2x+4x^2+8x^3+16x^4+32x^5) \\ &= 1-k^6 \\ \Rightarrow & (1+2x+4x^2+8x^3+16x^4+32x^5) = \frac{1-k^6}{(1-k)} \\ \Rightarrow & (1+2x+(2x)^2+(2x)^3+(2x)^4+(2x)^5) = \frac{1-k^6}{(1-k)} \\ \Rightarrow & \frac{1-(2x)^6}{1-2x} = \frac{1-k^6}{(1-k)} \\ \Rightarrow & k = 2x \\ \therefore & \left(\frac{k}{x}\right) = 2 \end{aligned}$$

64. Clearly, $\alpha + \beta = 3$, $\alpha\beta = a$

and $\gamma + \delta = 12$, $\gamma\delta = b$

Also, it is given that,

$\alpha, \beta, \gamma, \delta \in \text{GP}$

Let A be the first term and R be the common ratio.

$$\text{Now, } \frac{\gamma + \delta}{\alpha + \beta} = \frac{12}{3} = 4$$

$$\Rightarrow \frac{AR^2 + AR^3}{A + AR} = 4$$

$$\Rightarrow \frac{AR^2(1+R)}{A(1+R)} = 4$$

$$\Rightarrow R^2 = 4$$

$$\therefore R = 2$$

$$\text{and } A = 1$$

$$\text{Now, } a = \alpha\beta = A \cdot AR = A^2R = 2$$

$$b = \gamma\delta = AR^2 \cdot AR^3 = A^2R^5 = (2)^5 = 32$$

65. Since x, y and z are in GP, so $y = xr$ and $z = xr^2$

We have, $x + y + z = ax$

$$\Rightarrow x + xr + xr^2 = ax$$

$$\Rightarrow r^2 + r + (1-a) = 0$$

Since r is real, so $D \geq 0$

$$\therefore 1 - 4(1-a) \geq 0$$

$$\Rightarrow 1 - 4 + 4a \geq 0$$

$$\Rightarrow 4a \geq 3$$

$$\Rightarrow a \geq \frac{3}{4}$$

Hence, the value of a is $\left[\frac{3}{4}, \infty\right)$.

66. Do yourself.

67. Do yourself.

68. Do yourself.

69. We have,

$$\begin{aligned} \text{Sum} &= (x+y) + (x^2+xy+y^2) \\ & \quad + (x^3+x^2y+xy^2+y^3) + \dots \text{ to } n \text{ terms} \\ &= \frac{1}{(x-y)} [(x^2-y^2) + (x^3-y^3) + \dots + (x^n-y^n)] \\ &= \frac{1}{(x-y)} [(x^2+x^3+\dots+x^n) - (y^2+y^3+\dots+y^n)] \\ &= \frac{1}{(x-y)} \left[\frac{x^2(1-x^n)}{(1-x)} - \frac{y^2(1-y^n)}{(1-y)} \right] \end{aligned}$$

70. We have,

$$\begin{aligned} \text{Sum} &= \frac{a}{(1+i)} + \frac{a}{(1+i)^2} + \frac{a}{(1+i)^3} + \dots + \frac{a}{(1+i)^n} \\ &= \frac{a}{(a+i)} \left[1 + \frac{1}{(1+i)} + \frac{1}{(1+i)^2} + \dots + \frac{1}{(1+i)^{n-1}} \right] \\ &= \frac{a}{(a+i)} \left(\frac{1 - \frac{1}{(1+i)^n}}{1 - \frac{1}{(1+i)}} \right) \end{aligned}$$

71. We have $\sum_{k=1}^{10} (2+3^k)$

$$\begin{aligned} &= (2+3) + (2+3^2) + (2+3^3) + \dots + (2+3^{10}) \\ &= 10 \times 2 + 3(1+3+3^2+\dots+3^9) \\ &= 20 + 3 \left(\frac{3^{10}-1}{3-1} \right) \\ &= 20 + \frac{3}{2} (3^{10}-1) \\ &= \left(\frac{37+3^{11}}{2} \right) \end{aligned}$$

72. We have,

$$\begin{aligned}
 & \sum_{n=1}^n (2^{n-1} + 3^n) \\
 &= \sum_{n=1}^n 2^{n-1} + \sum_{n=1}^n 3^n \\
 &= (1 + 2 + 2^2 + \dots + 2^{n-1}) + (3 + 3^2 + 3^3 + \dots + 3^n) \\
 &= \left(\frac{2^n - 1}{2 - 1} \right) + 3 \left(\frac{3^n - 1}{3 - 1} \right) \\
 &= (2^n - 1) + \frac{3}{2}(3^n - 1)
 \end{aligned}$$

73. (i) We have

$$\begin{aligned}
 & \sum_{k=1}^n (4^{k-1} + 5^{k+1}) \\
 &= \sum_{k=1}^n 4^{k-1} + \sum_{k=1}^n 5^{k+1} \\
 &= (1 + 4 + 4^2 + \dots + 4^{n-1}) \\
 &\quad + (5^2 + 5^3 + 5^4 + \dots + 5^{n+1}) \\
 &= \left(\frac{4^n - 1}{4 - 1} \right) + 5^2 \left(\frac{5^n - 1}{5 - 1} \right) \\
 &= \left(\frac{4^n - 1}{3} \right) + \frac{25}{4}(5^n - 1)
 \end{aligned}$$

(ii) Do yourself.

74. (i) We have,

$$\begin{aligned}
 & 5 + 55 + 555 + \dots \text{ to } n \text{ terms} \\
 &= 5(1 + 11 + 111 + \dots \text{ to } n \text{ terms}) \\
 &= \frac{5}{9}(9 + 99 + 999 + \dots \text{ to } n \text{ terms}) \\
 &= \frac{5}{9}[(10 - 1) + (10^2 - 1) + (10^3 - 1) + \dots + (10^n - 1)] \\
 &= \frac{5}{9}[10(1 + 10 + 10^2 + \dots + 10^{n-1}) - n] \\
 &= \frac{5}{9} \left[10 \left(\frac{10^n - 1}{10 - 1} \right) - n \right] \\
 &= \frac{5}{9} \left[\left(\frac{10^{n+1} - 10}{9} \right) - n \right] \\
 &= \frac{5}{81}(10^{n+1} - 9n - 10)
 \end{aligned}$$

(ii) Do yourself.

(iii) Do yourself.

75. We have,

$$\begin{aligned}
 \text{Sum} &= (6666 \dots 6)^2 + (888 \dots 8) \\
 &= 36(111 \dots 11)^2 + 8(111 \dots 11) \\
 &= 36(1 + 10 + 10^2 + \dots + 10^{n-1})^2 \\
 &\quad + 8(1 + 10 + 10^2 + \dots + 10^{n-1})
 \end{aligned}$$

$$\begin{aligned}
 &= 36 \left(\frac{10^n - 1}{10 - 1} \right)^2 + 8 \left(\frac{10^n - 1}{10 - 1} \right) \\
 &= 36 \left(\frac{10^n - 1}{9} \right)^2 + 8 \left(\frac{10^n - 1}{9} \right) \\
 &= \left(\frac{10^n - 1}{9} \right) \left(36 \left(\frac{10^n - 1}{9} \right) + 8 \right) \\
 &= \left(\frac{10^n - 1}{9} \right) [4(10^n - 1) + 8] \\
 &= \left(\frac{10^n - 1}{9} \right) [4(10^n - 1 + 2)] \\
 &= \frac{4}{9}(10^n - 1)(10^n + 1) \\
 &= \frac{4}{9}(10^{2n} - 1)
 \end{aligned}$$

76. We have,

$$\begin{aligned}
 \text{Sum} &= \sum_{n=1}^{10} \left[\left(\frac{1}{2} \right)^{n-1} + \left(\frac{1}{5} \right)^{n+1} \right] \\
 &= \sum_{n=1}^{10} \left[\left(\frac{1}{2} \right)^{n-1} \right] + \sum_{n=1}^{10} \left[\left(\frac{1}{5} \right)^{n+1} \right] \\
 &= \left(1 + \frac{1}{2} + \left(\frac{1}{2} \right)^2 + \dots + \left(\frac{1}{2} \right)^9 \right) \\
 &\quad + \left(\left(\frac{1}{5} \right)^2 + \left(\frac{1}{5} \right)^3 + \dots + \left(\frac{1}{5} \right)^{11} \right) \\
 &= \left(\frac{1 - \left(\frac{1}{2} \right)^{10}}{1 - \left(\frac{1}{2} \right)} \right) + \left(\frac{1}{5} \right)^2 \left(\frac{1 - \left(\frac{1}{5} \right)^{10}}{1 - \left(\frac{1}{5} \right)} \right) \\
 &= 2 \left(1 - \left(\frac{1}{2} \right)^{10} \right) + \left(\frac{1}{20} \right) \left(1 - \left(\frac{1}{5} \right)^{10} \right)
 \end{aligned}$$

77. We have,

$$\begin{aligned}
 \text{Sum} &= 11 + 103 + 1005 + \dots \\
 &= (10 + 1) + (10^2 + 3) + \dots + [10^n + (2n - 1)] \\
 &= (10 + 10^2 + \dots + 10^n) + [1 + 3 + 5 + \dots + (2n - 1)] \\
 &= 10(1 + 10 + \dots + 10^{n-1}) + [1 + 3 + 5 + \dots + (2n - 1)] \\
 &= 10 \left(\frac{10^n - 1}{10 - 1} \right) + n^2 \\
 &= \frac{10}{9}(10^n - 1) + n^2
 \end{aligned}$$

78. It is given that

$$\begin{aligned}
 & 1 + 3 + 3^2 + 3^3 + \dots + 3^{n-1} > 7000 \\
 \Rightarrow & \left(\frac{3^n - 1}{3 - 1} \right) > 7000 \\
 \Rightarrow & (3^n - 1) > 14000
 \end{aligned}$$

$$\Rightarrow 3^n > 14001$$

$$\Rightarrow 3^n > 3^9$$

$$\therefore n > 9$$

Hence, the least value of n is 9.

79. We have,

$$\begin{aligned} \text{Sum} &= \frac{1}{3} + \frac{5}{9} + \frac{19}{27} + \frac{65}{81} + \dots \text{ to } n \text{ terms} \\ &= \left(1 - \frac{2}{3}\right) + \left(1 - \left(\frac{2}{3}\right)^2\right) + \left(1 - \left(\frac{2}{3}\right)^3\right) \\ &\quad + \dots + \left(1 - \left(\frac{2}{3}\right)^n\right) \\ &= n - \frac{2}{3} \left(1 + \left(\frac{2}{3}\right) + \left(\frac{2}{3}\right)^2 + \dots + \left(\frac{2}{3}\right)^{n-1}\right) \\ &= n - \frac{2}{3} \left(\frac{1 - \left(\frac{2}{3}\right)^n}{1 - \frac{2}{3}}\right) \\ &= n - 2 \left(1 - \left(\frac{2}{3}\right)^n\right) \end{aligned}$$

80. We have,

$$\begin{aligned} S &= \frac{2}{3} + \frac{8}{9} + \frac{26}{27} + \frac{30}{81} + \dots \text{ to } n\text{-terms} \\ &= \left(1 - \frac{1}{3}\right) + \left(1 - \frac{1}{9}\right) + \left(1 - \frac{1}{27}\right) + \dots \\ &= \left(1 - \frac{1}{3}\right) + \left(1 - \left(\frac{1}{3}\right)^2\right) + \left(1 - \left(\frac{1}{3}\right)^3\right) \\ &\quad + \dots + \left(1 - \left(\frac{1}{3}\right)^n\right) \\ &= n - \frac{1}{3} \left(1 + \frac{1}{3} + \left(\frac{1}{3}\right)^2 + \dots + \left(\frac{1}{3}\right)^{n-1}\right) \\ &= n - \frac{1}{3} \left(\frac{1 - \left(\frac{1}{3}\right)^n}{1 - \left(\frac{1}{3}\right)}\right) \\ &= n - \frac{1}{2} \left(1 - \left(\frac{1}{3}\right)^n\right) \end{aligned}$$

81. We have,

$$\begin{aligned} S &= a + ar + ar^2 + \dots + ar^{n-1} = a \left(\frac{r^n - 1}{r - 1}\right) \\ P &= a \cdot ar \cdot ar^2 \dots ar^{n-1} \\ &= an \cdot r^{1+2+3+\dots+(n-1)} \\ &= a^n r^{\frac{n(n-1)}{2}} \end{aligned}$$

$$\begin{aligned} \text{and } R &= \frac{1}{a} + \frac{1}{ar} + \frac{1}{ar^2} + \dots + \frac{1}{ar^{n-1}} \\ &= \frac{1}{a} \left(1 + \frac{1}{r} + \frac{1}{r^2} + \dots + \frac{1}{r^{n-1}}\right) \\ &= \frac{1}{a} \left(\frac{1 - \left(\frac{1}{r}\right)^n}{1 - \frac{1}{r}}\right) \\ &= \frac{1}{a} \left(\frac{r^n - 1}{r^{n-1}(r - 1)}\right) \\ &= \frac{1}{ar^{n-1}} \left(\frac{r^n - 1}{r - 1}\right) \end{aligned}$$

$$\begin{aligned} \text{Now, } \left(\frac{S}{R}\right) &= \frac{a(r^n - 1)}{(r - 1)} + \frac{1}{ar^{n-1}} \left(\frac{r^n - 1}{r - 1}\right) \\ &= a^2 r^{n-1} \\ \left(\frac{S}{R}\right)^n &= a^{2n} r^{n(n-1)} = \left(a^n r^{\frac{n(n-1)}{2}}\right)^2 = P^2 \end{aligned}$$

Hence, the result.

82. Clearly, $f(1) = 3, f(2) = 3^2, f(3) = 3^3, \dots, f(n) = 3^n$

$$\text{Now, } \sum_{x=1}^n f(x) = 120$$

$$\Rightarrow f(1) + f(2) + f(3) + \dots + f(n) = 120$$

$$\Rightarrow 3 + 3^2 + 3^3 + \dots + 3^n = 120$$

$$\Rightarrow 3 \left(\frac{3^n - 1}{3 - 1}\right) = 120$$

$$\Rightarrow \left(\frac{3^n - 1}{3 - 1}\right) = 40$$

$$\Rightarrow 3^n - 1 = 80$$

$$\Rightarrow 3^n = 81 = 3^4$$

$$\therefore n = 4$$

Hence, the value of n is 4.

83. We have,

$$\sum_{n=1}^{100} a_{2n} = \alpha$$

$$\begin{aligned} \Rightarrow \alpha &= a_2 + a_4 + a_6 + \dots + a_{200} \\ &= ar + ar^3 + ar^5 + \dots + ar^{199} \\ &= ar(1 + r^2 + r^4 + \dots + r^{198}) \end{aligned} \quad \dots(i)$$

$$\text{Also, } \sum_{n=1}^{100} a_{2n-1} = \beta,$$

$$\begin{aligned} \Rightarrow \beta &= a_1 + a_3 + a_5 + \dots + a_{199} \\ &= a + ar^2 + ar^4 + \dots + ar^{198} \\ &= a(1 + r^2 + r^4 + \dots + r^{198}) \end{aligned}$$

$$\text{Now, } \frac{\alpha}{\beta} = \frac{ar(1+r^2+r^4+\dots+r^{198})}{(1+r^2+r^4+\dots+r^{198})}$$

$$\Rightarrow r = \frac{\alpha}{\beta}$$

Hence, the result.

84. We have,

$$\begin{aligned} \text{Sum} &= \frac{(\sqrt{2}+1)}{1-\left(\frac{1}{\sqrt{2}+1}\right)} \\ &= \frac{(\sqrt{2}+1)}{1-(\sqrt{2}-1)} \\ &= \frac{(\sqrt{2}+1)}{2-\sqrt{2}} \\ &= \frac{(\sqrt{2}+1)}{\sqrt{2}(\sqrt{2}-1)} \\ &= \frac{(\sqrt{2}+1)}{\sqrt{2}} = \frac{(\sqrt{2}+1)^2}{\sqrt{2}} = \left(\frac{3+2\sqrt{2}}{\sqrt{2}}\right) \end{aligned}$$

85. We have,

$$\begin{aligned} \text{Sum} &= \frac{1}{2} + \frac{1}{3^2} + \frac{1}{2^3} + \frac{1}{3^4} + \frac{1}{2^5} + \frac{1}{3^6} + \dots \infty \\ &= \left(\frac{1}{2} + \frac{1}{2^3} + \frac{1}{2^5} + \dots\right) + \left(\frac{1}{3^2} + \frac{1}{3^4} + \frac{1}{3^6} + \dots\right) \\ &= \left(\frac{\frac{1}{2}}{1-\frac{1}{2^2}}\right) + \left(\frac{\frac{1}{3^2}}{1-\frac{1}{3^2}}\right) \\ &= \frac{2}{3} + \frac{1}{8} = \frac{19}{24} \end{aligned}$$

86. We have,

$$\begin{aligned} b &= a + a^2 + a^3 + \dots \infty \\ \Rightarrow b &= \frac{a}{1-a} \\ \Rightarrow b - ab &= a \\ \Rightarrow a(1+b) &= b \\ \Rightarrow a &= \frac{b}{(1+b)} \end{aligned}$$

Hence, the result.

87. We have,

$$\begin{aligned} x &= a + \frac{a}{r} + \frac{a}{r^2} + \dots \infty \\ \Rightarrow x &= \frac{a}{1-\frac{1}{r}} = \frac{ar}{r-1} \end{aligned}$$

Similarly,

$$y = \frac{b}{1+\frac{1}{r}} = \frac{br}{r+1}$$

$$\text{and } z = \frac{c}{1-\frac{1}{r^2}} = \frac{cr^2}{r^2-1}$$

$$\text{Now, } \frac{xy}{z} = \frac{\frac{abr^2}{r^2-1}}{\frac{cr^2}{r^2-1}} = \frac{ab}{c}$$

88. We have,

$$\begin{aligned} x &= 1 + a + a^2 + \dots \infty \\ \Rightarrow x &= \frac{1}{1-a} \\ \Rightarrow 1-a &= \frac{1}{x} \\ \Rightarrow a &= \frac{x-1}{x} \end{aligned}$$

$$\text{Similarly, } b = \frac{y-1}{y}$$

Now,

$$\begin{aligned} 1 + ab + (ab)^2 + (ab)^3 + \dots \\ &= \frac{1}{1-ab} \\ &= \frac{1}{1-\left(\frac{x-1}{x}\right)\left(\frac{y-1}{y}\right)} \\ &= \frac{xy}{xy - (xy - x - y + 1)} \\ &= \frac{xy}{x+y-1} \end{aligned}$$

89. We have,

$$\begin{aligned} A &= 1 + r^a + r^{2a} + \dots \text{ to } \infty \\ \Rightarrow A &= \frac{1}{1-r^a} \\ \Rightarrow (1-r^a) &= \frac{1}{A} \\ \Rightarrow r^a &= \left(1 - \frac{1}{A}\right) \\ \Rightarrow r &= \left(\frac{A-1}{A}\right)^{1/a} \end{aligned}$$

$$\text{Similarly, } r = \left(\frac{B-1}{B}\right)^{1/b}$$

$$\text{Hence, } \left(\frac{A-1}{A}\right)^{1/a} = r = \left(\frac{B-1}{B}\right)^{1/b}$$

90. We have,

$$\begin{aligned} x &= \sum_{n=0}^{\infty} \cos^{2n}\theta \\ &= 1 + \cos^2\theta + \cos^4\theta + \cos^6\theta + \dots \\ &= \frac{1}{1-\cos^2\theta} = \frac{1}{\sin^2\theta} \end{aligned}$$

$$y = \sum_{n=0}^{\infty} \sin^{2n} \varphi$$

$$= 1 + \sin^2 \varphi + \sin^4 \varphi + \sin^6 \varphi + \dots$$

$$= \frac{1}{1 - \sin^2 \varphi} = \frac{1}{\cos^2 \varphi}$$

and $z = \sum_{n=0}^{\infty} \cos^{2n} \theta \sin^{2n} \varphi$

$$= 1 + \cos^2 \theta \sin^2 \varphi + \cos^4 \theta \sin^4 \varphi + \dots + \cos^6 \theta \sin^6 \varphi + \dots$$

$$= \frac{1}{1 - \cos^2 \theta \sin^2 \varphi} = \frac{1}{1 - (1 - \sin^2 \theta)(1 - \cos^2 \varphi)}$$

$$= \frac{1}{1 - \left(1 - \frac{1}{x}\right)\left(1 - \frac{1}{y}\right)}$$

$$= \frac{xy}{xy - (x-1)(y-1)}$$

$$= \frac{xy}{xy - (xy - x - y + 1)} = \frac{xy}{x + y - 1}$$

$$xz + yz - z = xy$$

91. We have,

$$\text{Sum} = (x+y) + (x^2+xy+y^2) + (x^3+x^2y+xy^2+y^3) + \dots \text{ to } \infty$$

$$= \frac{1}{x-y} [(x^2-y^2) + (x^3-y^3) + (x^4-y^4) + \dots]$$

$$= \frac{1}{x-y} [(x^2+x^3+x^4+\dots) - (y^2+y^3+y^4+\dots)]$$

$$= \frac{1}{(x-y)} \left(\frac{x^2}{1-x} - \frac{y^2}{1-y} \right)$$

92. We have,

$$S_p = 1 + r^p + r^{2p} + \dots$$

$$= \frac{1}{1 - r^p}$$

$$\text{Also, } s_p = 1 - r^p - r^{2p} - \dots$$

$$= \frac{1}{1 - (-r^p)} = \frac{1}{1 + r^p}$$

Now,

$$S_p + s_p = \frac{1}{1 - r^p} + \frac{1}{1 + r^p}$$

$$= \frac{1 + r^p + 1 - r^p}{(1 - r^p)(1 + r^p)}$$

$$= \frac{2}{(1 - r^{2p})}$$

$$= 2S_{2p}$$

93. We have,

$$8^{1+|\cos x|+|\cos x|^2+|\cos x|^3+\dots \text{ to } \infty} = 4^3$$

$$\Rightarrow 8^{\frac{1}{1-|\cos x|}} = 4^3 = 64 = 8^2$$

$$\Rightarrow \frac{1}{1-|\cos x|} = 2$$

$$\Rightarrow |1-|\cos x|| = \frac{1}{2}$$

$$\Rightarrow |\cos x| = \frac{1}{2}$$

$$\Rightarrow x = -\frac{2\pi}{3}, -\frac{\pi}{3}, \frac{\pi}{3}, \frac{2\pi}{3}$$

Hence, the solutions are

$$\left\{ \pm \frac{2\pi}{3}, \pm \frac{\pi}{3} \right\}$$

94. Given equation is

$$x^2 - 9x + 8 = 0$$

$$\Rightarrow (x-1)(x-8) = 0$$

$$\Rightarrow x = 1 \text{ or } 8$$

Now, given expression is

$$\exp\{(\sin^2 x + \sin^4 x + \sin^6 x + \dots \infty) \log_e 2\}$$

$$= 2^{\sin^2 x + \sin^4 x + \sin^6 x + \dots}$$

$$= 2^{\frac{\sin^2 x}{1 - \sin^2 x}} = 2^{\tan^2 x}$$

$$\text{When, } 2^{\tan^2 x} = 1 = 2^0$$

$$\Rightarrow \tan^2 x = 0$$

$$\Rightarrow \tan x = 0$$

$$\text{When } 2^{\tan^2 x} = 8 = 2^3$$

$$\Rightarrow \tan^2 x = 3 = (\sqrt{3})^2$$

$$\Rightarrow \tan x = \sqrt{3}$$

$$\text{Now, } \frac{\cos x}{\cos x + \sin x} = \frac{1}{1 + \tan x} = \frac{1}{\sqrt{3} + 1}$$

$$= \frac{(\sqrt{3} - 1)}{2}$$

$$95. \text{ Let } S = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

$$= \frac{1}{4} \left(1 + \frac{1}{2} + \frac{1}{4} + \dots \right)$$

$$= \frac{1}{4} \times \frac{1}{1 - \frac{1}{2}} = \frac{1}{4} \times 2 = \frac{1}{2}$$

We have,

$$(0.2)^{\log_{\sqrt{2}} \left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \right)} = (0.2)^{\log_{\sqrt{2}} \left(\frac{1}{2} \right)}$$

$$= (0.2)^{\log_{2^{1/2}} (2^{-1})}$$

$$= (0.2)^{-2 \log_2 2}$$

$$= (0.2)^{-2} = \left(\frac{1}{5} \right)^{-2} = 5^2 = 25$$

96. Given p, q and r are in AP,

$$\Rightarrow 2q = p + r$$

Let a be the first term and R be the common ratio of the GP.

$$\text{Now, } t_p = aR^{p-1}, t_q = aR^{q-1}, t_r = aR^{r-1}$$

$$\text{Here, } \frac{t_q}{t_p} = \frac{aR^{q-1}}{aR^{p-1}} = R^{q-p}$$

$$\text{and } \frac{t_r}{t_q} = \frac{aR^{r-1}}{aR^{q-1}} = R^{r-q} = R^{q-p}$$

Thus, t_p, t_q, t_r are in GP.

97. It is given that a, b and c are in AP.

$$2b = a + c$$

Also, x, y and z are in GP.

$$y^2 = xz$$

$$y = \sqrt{xz}$$

Now,

$$\begin{aligned} x^{b-c} \cdot y^{c-a} \cdot z^{a-b} &= x^{b-c} \cdot (\sqrt{xz})^{c-a} \cdot z^{a-b} \\ &= x^{b-c} \cdot (xz)^{\frac{c-a}{2}} \cdot z^{a-b} \\ &= x^{b-c+\frac{c-a}{2}} \cdot z^{a-b+\frac{c-a}{2}} \\ &= x^{\frac{2b-2c+c-a}{2}} \cdot z^{\frac{2a-2b+c-a}{2}} \\ &= x^{\frac{2b-c-a}{2}} \cdot z^{\frac{a+c-2b}{2}} \\ &= x^{\frac{2b-2b}{2}} \cdot z^{\frac{2b-2b}{2}} \\ &= x^0 \cdot z^0 = 1 \end{aligned}$$

98. It is given that a, b and c are in GP.

$$\therefore b^2 = ac$$

$$\Rightarrow \log_n(b^2) = \log_n(ac)$$

$$\Rightarrow 2 \log_n(b) = \log_n(a) + \log_n(c)$$

$$\Rightarrow \log_n(a), \log_n(b), \log_n(c) \text{ are in AP.}$$

99. It is given that a, b and c are in GP.

$$\Rightarrow b^2 = ac$$

$$\text{Also, } x = \frac{a+b}{2}, y = \frac{b+c}{2}$$

Now,

$$\begin{aligned} \frac{a}{x} + \frac{c}{y} &= \frac{2a}{a+b} + \frac{2c}{b+c} \\ &= \frac{2(ab+ac+ac+bc)}{(a+b)(b+c)} \\ &= \frac{2(ab+2b^2+bc)}{(ab+ac+b^2+bc)} \\ &= \frac{2(ab+2b^2+bc)}{(ab+2b^2+bc)} \\ &= 2 \end{aligned}$$

$$\begin{aligned} \text{Also, } \frac{1}{x} + \frac{1}{y} &= \frac{2}{(a+b)} + \frac{2}{(b+c)} \\ &= \frac{2(a+b+b+c)}{(a+b)(b+c)} \\ &= \frac{2(a+2b+c)}{(ab+ac+b^2+bc)} \\ &= \frac{2(a+2b+c)}{(ab+b^2+b^2+bc)} \\ &= \frac{2(a+2b+c)}{(ab+2b^2+bc)} \\ &= \frac{2(a+2b+c)}{b(a+2b+c)} = \frac{2}{b} \end{aligned}$$

Hence, the result.

100. We have a, b and $3c$ are in AP and a, b and $4c$ are in GP.

$$\therefore 2b = a + 3c \text{ and } b^2 = 4ac$$

Squaring $2b = a + 3c$, we get

$$\Rightarrow 4b^2 = a^2 + 9c^2 + 6ac$$

$$\Rightarrow 16ac = a^2 + 9c^2 + 6ac \quad (\because b^2 = 4ac)$$

$$\Rightarrow a^2 + 9c^2 - 10ac = 0$$

$$\Rightarrow a^2 - 10ac + 9c^2 = 0$$

$$\Rightarrow (a-c)(a-9c) = 0$$

$$\Rightarrow a = c \text{ or } 9c$$

When $a = c$,

$$b = \left(\frac{a+3c}{2} \right) = \left(\frac{c+3c}{2} \right) = 2c$$

$$\Rightarrow \frac{a}{b} = \frac{c}{2c} = \frac{1}{2}$$

When $a = 9c$

$$\Rightarrow b = \left(\frac{a+3c}{2} \right) = \left(\frac{9c+3c}{2} \right) = 6c$$

$$\Rightarrow \frac{a}{b} = \frac{9c}{6c} = \frac{3}{2}$$

101. Given $x^a = x^{\frac{b}{2}} z^{\frac{b}{2}} = z^c$

$$\Rightarrow x^a = x^{\frac{b}{2}} z^{\frac{b}{2}} = z^c = k \text{ (say)}$$

$$\Rightarrow x = k^{\frac{1}{a}}, z = k^{\frac{1}{c}}, (xz) = k^{\frac{2}{b}}$$

$$\text{Now, } (xz) = k^{\frac{2}{b}}$$

$$\Rightarrow (k^{\frac{1}{a}} k^{\frac{1}{c}}) = k^{\frac{2}{b}}$$

$$\Rightarrow k^{\frac{1}{a} + \frac{1}{c}} = k^{\frac{2}{b}}$$

$$\Rightarrow \frac{1}{a} + \frac{1}{c} = \frac{2}{b}$$

$$\Rightarrow \frac{1}{a}, \frac{1}{b}, \frac{1}{c} \in \text{AP}$$

Hence, the result.

102. We have,

$$\begin{aligned} a + b + c &= xb \\ \Rightarrow a + ar + ar^2 &= arx \\ \Rightarrow 1 + r + r^2 &= rx \\ \Rightarrow r^2 + (1-x)r + 1 &= 0 \end{aligned}$$

Since a, b and c are three distinct real numbers, so its $D > 0$

$$\begin{aligned} \Rightarrow (1-x)^2 - 4 &> 0 \\ \Rightarrow (x-1)^2 - (2)^2 &> 0 \\ \Rightarrow (x-1+2)(x-1-2) &> 0 \\ \Rightarrow (x+1)(x-3) &> 0 \\ \Rightarrow x < -1 \text{ or } x > 3 \end{aligned}$$

103. Do yourself.

104. Given a, b and c are in AP.

$$\Rightarrow 2b = a + c \quad \dots(i)$$

Also, a^2, b^2 and c^2 are in GP.

$$\Rightarrow b^4 = a^2 c^2 \quad \dots(ii)$$

$$\Rightarrow a + b + c = \frac{3}{2}$$

$$\Rightarrow 3b = \frac{3}{2}$$

$$\Rightarrow b = \frac{1}{2}$$

From Eq. (ii), we get,

$$a^2 c^2 = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

$$\Rightarrow ac = \pm \frac{1}{4}$$

Also $a + c = 2b = 1$

Thus, a and c are the roots of $x^2 - x \pm \frac{1}{4} = 0$.

$$\Rightarrow x^2 - x + \frac{1}{4} = 0 \text{ and } x^2 - x - \frac{1}{4} = 0$$

$$\Rightarrow \left(x - \frac{1}{2}\right)^2 = 0 \text{ and } \left(x - \frac{1}{2}\right)^2 = \frac{1}{2}$$

$$\Rightarrow x = \frac{1}{2} \text{ and } x = \frac{1}{2} \pm \frac{1}{\sqrt{2}}$$

$$\Rightarrow x = \frac{1}{2} \pm \frac{1}{\sqrt{2}}, \text{ since } a < b < c.$$

$$\text{Thus, } a = \frac{1}{2} - \frac{1}{\sqrt{2}}, \text{ as } a < b < c.$$

105. We have

$$\frac{a^{n+1} + b^{n+1}}{a^n + b^n} = \sqrt{ab}$$

$$\Rightarrow \frac{a^{n+1} + b^{n+1}}{a^n + b^n} = (ab)^{1/2}$$

$$\Rightarrow a^{n+1} + b^{n+1} = a^{n+\frac{1}{2}} b^{\frac{1}{2}} + b^{n+\frac{1}{2}} a^{\frac{1}{2}}$$

$$\Rightarrow a^{n+\frac{1}{2}} (a^{\frac{1}{2}} - b^{\frac{1}{2}}) = b^{n+\frac{1}{2}} (a^{\frac{1}{2}} - b^{\frac{1}{2}})$$

$$\Rightarrow a^{n+\frac{1}{2}} = b^{n+\frac{1}{2}}$$

$$\Rightarrow \left(\frac{a}{b}\right)^{n+\frac{1}{2}} = 1 = \left(\frac{a}{b}\right)^0$$

$$\Rightarrow n + \frac{1}{2} = 0$$

$$\Rightarrow n = -\frac{1}{2}$$

106. Do yourself.

107. Let a and b be two positive numbers.

$$\therefore x = \frac{a+b}{2}$$

and $a, y, z, b \in \text{GP}$

$$\Rightarrow t_4 = b$$

$$\Rightarrow ar^3 = b$$

$$\Rightarrow r^3 = \frac{b}{a}$$

$$\Rightarrow r = \left(\frac{b}{a}\right)^{1/3}$$

$$\text{Now, } y = ar = a \left(\frac{b}{a}\right)^{1/3} = a^{2/3} b^{1/3}$$

$$\text{and } z = ar^2 = a \left(\frac{b}{a}\right)^{2/3} = a^{1/3} b^{2/3}$$

Therefore, $y^3 + z^3 = a^2 b + ab^2 = ab(a+b)$

$$\Rightarrow y^3 + z^3 = 2xyz$$

$$\Rightarrow \frac{y^3 + z^3}{xyz} = 2 \dots$$

Hence, the result.

108. Do yourself

109. If one geometric mean G and two arithmetic means A_1 and A_2 be inserted between two given quantities, prove that $G^2 = (2A_1 - A_2)(2A_2 - A_1)$.

110. Given a, b, c are in GP

$$\Rightarrow b^2 = ac$$

$$\Rightarrow b = \sqrt{ac}$$

Given equation is $ax^2 + 2bx + c = 0$

$$\Rightarrow ax^2 + 2\sqrt{ac}x + c = 0$$

$$\Rightarrow (\sqrt{ax} + \sqrt{c})^2 = 0$$

$$\Rightarrow (\sqrt{ax} + \sqrt{c}) = 0$$

$$\Rightarrow x = -\frac{\sqrt{c}}{\sqrt{a}}$$

Since the equations have a common root, so

$$d\left(\frac{c}{a}\right) - 2e\left(\frac{\sqrt{c}}{\sqrt{a}}\right) + f = 0$$

$$\Rightarrow d\left(\frac{c}{a}\right) - 2e\left(\frac{\sqrt{c^2}}{\sqrt{ac}}\right) + f = 0$$

$$\Rightarrow d\left(\frac{c}{a}\right) - 2e\left(\frac{c}{b}\right) + f = 0$$

$$\Rightarrow \left(\frac{d}{a}\right) - 2\left(\frac{e}{b}\right) + \frac{f}{c} = 0$$

$$\Rightarrow \left(\frac{d}{a}\right) + \frac{f}{c} = 2\left(\frac{e}{b}\right)$$

$$\Rightarrow \frac{d}{a}, \frac{e}{b}, \frac{f}{c} \text{ are in AP}$$

111. Let two numbers be a and b respectively. It is given that,

$$a + b = 6\sqrt{ab}$$

$$\Rightarrow (a + b)^2 = 36ab$$

$$\Rightarrow (a - b)^2 = (a + b)^2 - 4ab = 32ab$$

$$\text{Now, } \frac{(a + b)^2}{(a - b)^2} = \frac{36ab}{32ab}$$

$$\Rightarrow \frac{(a + b)^2}{(a - b)^2} = \frac{9}{8}$$

$$\Rightarrow \frac{(a + b)}{(a - b)} = \frac{3}{2\sqrt{2}}$$

$$\Rightarrow \frac{(a + b) + (a - b)}{(a + b) - (a - b)} = \frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}}$$

$$\Rightarrow \frac{2a}{2b} = \frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}}$$

$$\Rightarrow \frac{a}{b} = \frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}}$$

112. In AP, $t_m = \frac{1}{n} \Rightarrow a + (m - 1)d = \frac{1}{n}$

$$\text{and } t_n = \frac{1}{m} \Rightarrow a + (n - 1)d = \frac{1}{m}$$

On subtraction, we get,

$$(m - n)d = \frac{1}{n} - \frac{1}{m} = \frac{(m - n)}{mn}$$

$$\Rightarrow d = \frac{1}{mn}$$

Put $d = \frac{1}{mn}$ in (i), we get,

$$a + (m - 1)\frac{1}{mn} = \frac{1}{n}$$

$$\Rightarrow a + \frac{1}{n} - \frac{1}{mn} = \frac{1}{n}$$

$$\Rightarrow a = \frac{1}{mn}$$

$$\text{Thus, } t_{mn} = a + (mn - 1)d$$

$$= \frac{1}{mn} + (mn - 1)\frac{1}{mn}$$

$$= 1$$

Hence, the m th term of HP is

$$= \frac{1}{t_{mn}} = \frac{1}{1} = 1$$

113. In an AP,

$$t_m = \frac{1}{n} \Rightarrow a + (m - 1)d = \frac{1}{n}$$

$$\text{and } t_n = \frac{1}{m} \Rightarrow a + (n - 1)d = \frac{1}{m}$$

On subtraction, we get,

$$(m - n)d = \frac{1}{n} - \frac{1}{m} = \frac{(m - n)}{mn}$$

$$\Rightarrow d = \frac{1}{mn}$$

Put $d = \frac{1}{mn}$ in Eq. (i), we get

$$a + (m - 1)\frac{1}{mn} = \frac{1}{n}$$

$$\Rightarrow a + \frac{1}{n} - \frac{1}{mn} = \frac{1}{n}$$

$$\Rightarrow a = \frac{1}{mn}$$

Thus, $t_{m+n} = a + (m + n - 1)d$

$$= \frac{1}{mn} + \frac{(m + n - 1)}{mn}$$

$$= \frac{1}{mn} + \frac{(m + n)}{mn} - \frac{1}{mn}$$

$$= \left(\frac{m + n}{mn}\right)$$

Hence, the $(m + n)$ th term is $\left(\frac{mn}{m + n}\right)$.

114. In an AP,

$$t_p = q \Rightarrow a + (p - 1)d = q$$

$$\text{and } t_q = p \Rightarrow a + (q - 1)d = p$$

On subtraction, we get

$$(p - q)d = (q - p)$$

$$\Rightarrow d = -1$$

Put $d = -1$, in Eq. (i), we get

$$\Rightarrow a - (p - 1) = q$$

$$\Rightarrow a = p + q - 1$$

Now,

$$t_n = a + (n - 1)d$$

$$= p + q - 1 - (n - 1)$$

$$= p + q - n$$

Hence, the n th term of the HP is

$$\frac{1}{t_n} = \frac{1}{p + q - n}$$

115. It is given that,

$$a_1, a_2, a_3 \in \text{AP} \Rightarrow 2a_2 = a_1 + a_3$$

$$\text{Also } a_2, a_3, a_4 \in \text{GP} \Rightarrow a_3^2 = a_2 \cdot a_4$$

$$\text{and, } a_3, a_4, a_5 \in \text{HP} \Rightarrow a_4 = \frac{2a_3a_5}{a_3 + a_5}$$

Now,

$$a_3^2 = a_2 \cdot a_4$$

$$\Rightarrow a_3^2 = \left(\frac{a_1 + a_3}{2} \right) \times \left(\frac{2a_3a_5}{a_3 + a_5} \right)$$

$$\Rightarrow a_3^2 = (a_1 + a_3) \times \left(\frac{a_3a_5}{a_3 + a_5} \right)$$

$$\Rightarrow a_3^2(a_3 + a_5) = a_3a_5(a_1 + a_3)$$

$$\Rightarrow a_3^3 + a_3^2a_5 = a_1a_3a_5 + a_3^2a_5$$

$$\Rightarrow a_3^3 = a_1a_3a_5$$

$$\Rightarrow a_3^2 = a_1a_5$$

$$\Rightarrow a_1, a_3, a_5 \in \text{GP}$$

116. $a^x = b^y = c^z = d^w = k$ (say)

It is given that a, b, c and d are in GP.

$$\therefore \frac{a}{b} = \frac{b}{c} = \frac{c}{d}$$

$$\Rightarrow \frac{k^{\frac{1}{x}}}{k^{\frac{1}{y}}} = \frac{k^{\frac{1}{y}}}{k^{\frac{1}{z}}} = \frac{k^{\frac{1}{z}}}{k^{\frac{1}{w}}}$$

$$\Rightarrow k^{\frac{1}{x} - \frac{1}{y}} = k^{\frac{1}{y} - \frac{1}{z}} = k^{\frac{1}{z} - \frac{1}{w}}$$

$$\Rightarrow \frac{1}{x} - \frac{1}{y} = \frac{1}{y} - \frac{1}{z} = \frac{1}{z} - \frac{1}{w}$$

$$\Rightarrow \frac{1}{x}, \frac{1}{y}, \frac{1}{z}, \frac{1}{w} \in \text{AP}$$

$$\Rightarrow x, y, z \text{ and } w \in \text{HP}$$

117. It is given that x, y and z are in GP.

$$\therefore y^2 = xz$$

$$\Rightarrow \log(y^2) = \log(xz)$$

$$\Rightarrow 2\log(y) = \log(x) + \log(z)$$

$$\Rightarrow \log(x), \log(y), \log(z) \in \text{AP}$$

$$\Rightarrow 1 + \log(x), 1 + \log(y), 1 + \log(z) \in \text{AP}$$

$$\Rightarrow \frac{1}{1 + \log(x)}, \frac{1}{1 + \log(y)}, \frac{1}{1 + \log(z)} \in \text{HP}$$

118. In AP, $t_p = \frac{1}{a} \Rightarrow A + (p-1)d = \frac{1}{a}$ (i)

$t_q = \frac{1}{b} \Rightarrow A + (q-1)d = \frac{1}{b}$ (ii)

and $t_r = \frac{1}{c} \Rightarrow A + (r-1)d = \frac{1}{c}$ (iii)

Solving Eqs (i), (ii) and (iii), we get

$$(p-q) = \frac{b-a}{abd}$$

$$(q-r) = \frac{(b-c)}{bcd}$$

and $(r-p) = \frac{(c-a)}{acd}$

Thus,

$$\begin{aligned} & \frac{q-r}{a} + \frac{r-p}{b} + \frac{p-q}{c} \\ &= \frac{(b-c)}{bcd} + \frac{(c-a)}{acd} + \frac{(a-b)}{abd} \\ &= \frac{1}{d} \left(\frac{1}{c} - \frac{1}{b} + \frac{1}{a} - \frac{1}{c} + \frac{1}{b} - \frac{1}{a} \right) \\ &= \frac{1}{d} \times 0 \\ &= 0. \end{aligned}$$

119. It is given that

$x_1, x_2, x_3, \dots, x_n$ are in HP.

$$\therefore \frac{1}{x_1}, \frac{1}{x_2}, \frac{1}{x_3}, \dots, \frac{1}{x_n} \in \text{AP}$$

$$\Rightarrow \frac{1}{x_2} - \frac{1}{x_1} = \frac{1}{x_3} - \frac{1}{x_2} = \dots = \frac{1}{x_n} - \frac{1}{x_{n-1}} = d$$

$$\Rightarrow \frac{x_1 - x_2}{x_1x_2} = \frac{x_2 - x_3}{x_2x_3} = \dots = \frac{x_{n-1} - x_n}{x_{n-1}x_n} = d$$

Now,

$$\begin{aligned} & x_1x_2 + x_2x_3 + \dots + x_{n-1}x_n \\ &= \frac{x_1 - x_2}{d} + \frac{x_2 - x_3}{d} + \dots + \frac{x_{n-1} - x_n}{d} \\ &= \frac{1}{d} (x_1 - x_2 + x_2 - x_3 + x_{n-1} - x_n) \\ &= \frac{1}{d} (x_1 - x_n) \\ &= (n-1)x_1x_n \end{aligned}$$

120. It is given that a, b and c are in AP.

$$\therefore 2b = a + c$$

It is given that

$$x = \sum_{n=0}^{\infty} a^n, y = \sum_{n=0}^{\infty} b^n, z = \sum_{n=0}^{\infty} c^n$$

$$\Rightarrow x = \frac{1}{1-a}, y = \frac{1}{1-b}, z = \frac{1}{1-c}$$

$$\Rightarrow (1-a) = \frac{1}{x}, (1-b) = \frac{1}{y}, (1-c) = \frac{1}{z}$$

$$\Rightarrow a = \left(1 - \frac{1}{x}\right), b = \left(1 - \frac{1}{y}\right), c = \left(1 - \frac{1}{z}\right)$$

$$\Rightarrow a = \left(\frac{x-1}{x}\right), b = \left(\frac{y-1}{y}\right), c = \left(\frac{z-1}{z}\right)$$

Now, putting these values in

$$2b = a + c, \text{ we have,}$$

$$2\left(\frac{y-1}{y}\right) = \left(\frac{x-1}{x}\right) + \left(\frac{z-1}{z}\right)$$

$$\Rightarrow 2\left(1 - \frac{1}{y}\right) = \left(1 - \frac{1}{x}\right) + \left(1 - \frac{1}{z}\right)$$

$$\Rightarrow \frac{2}{y} = \frac{1}{x} + \frac{1}{z}$$

$$\Rightarrow \frac{1}{x}, \frac{1}{y}, \frac{1}{z} \in \text{AP}$$

$$x, y, z \in \text{HP}$$

Hence, the result.

121. Given $a_1 = 2$. So,

$$a_{10} = a_1 + 9d$$

$$\Rightarrow 9d = a_{10} - a_1 = 3 - 2 = 1$$

$$\Rightarrow d = \frac{1}{9}$$

$$\text{Now, } a_4 = a_1 + 3d = 2 + \frac{1}{3} = \frac{7}{3}$$

$$\text{Also, } h_{10} = 3$$

$$\Rightarrow \frac{1}{h_{10}} = \frac{1}{3}$$

$$\Rightarrow \frac{1}{h_1} + 9D = \frac{1}{3}$$

$$\Rightarrow 9D = \frac{1}{3} - \frac{1}{h_1}$$

$$\Rightarrow 9D = \frac{1}{3} - \frac{1}{2} = -\frac{1}{6}$$

$$\Rightarrow D = -\frac{1}{54}$$

$$\text{Now, } \frac{1}{h_7} = \frac{1}{h_1} + 6D$$

$$\Rightarrow \frac{1}{h_7} = \frac{1}{2} - \frac{1}{9} = \frac{9-2}{18} = \frac{7}{18}$$

$$\Rightarrow h_7 = \frac{18}{7}$$

$$\text{Thus, } a_4 h_7 = \frac{7}{3} \times \frac{18}{7} = 6$$

$$\Rightarrow a_4 h_7 + 2007 = 6 + 2007 = 2013.$$

122. It is given that

$$a^2 + 9b^2 + 25c^2 = abc \left(\frac{15}{a} + \frac{5}{b} + \frac{3}{c} \right)$$

$$\Rightarrow (a)^2 + (3b)^2 + (5c)^2 = 15bc + 5ac + 3ab$$

$$\Rightarrow (a)^2 + (3b)^2 + (5c)^2 = (3b)(5c) + a(5c) + a(3b)$$

$$\Rightarrow (a)^2 + (3b)^2 + (5c)^2 - (3b)(5c) - a(5c) - a(3b) = 0$$

$$\Rightarrow \frac{1}{2}[(a-3b)^2 + (3b-5c)^2 + (5c-a)^2] = 0$$

$$\Rightarrow |(a-3b)^2 + (3b-5c)^2 + (5c-a)^2| = 0$$

$$\Rightarrow (a-3b)^2 = 0, (3b-5c)^2 = 0, (5c-a)^2 = 0$$

$$\Rightarrow (a-3b) = 0, (3b-5c) = 0, (5c-a) = 0$$

$$\Rightarrow a = 3b = 5c$$

$$\Rightarrow \frac{a}{1} = \frac{b}{3} = \frac{c}{5}$$

Clearly, a, b and $c \in \text{HP}$.

123. It is given that,

$$a, x \text{ and } b \text{ are in AP} \Rightarrow 2x = a + b$$

$$a, y \text{ and } b \in \text{GP} \Rightarrow y^2 = ab$$

$$a, z \text{ and } b \in \text{HP} \Rightarrow z = \frac{2ab}{a+b}$$

$$\text{Also, } x = 9z$$

$$\Rightarrow \frac{a+b}{2} = 9 \left(\frac{2ab}{a+b} \right)$$

$$\Rightarrow (a+b)(a+b) = 36ab$$

$$\Rightarrow (a+b) = 6\sqrt{ab} = 6|y|$$

$$\text{Clearly, } 2x = 6|y|$$

$$\Rightarrow x = 3|y|$$

$$\text{Also, } z = \frac{2ab}{a+b} = \frac{2y^2}{6|y|} = \frac{|y|}{3}$$

$$\Rightarrow |y| = 3z$$

Hence, the result.

124. Do yourself.

125. It is given that

$$4a^2 + 9b^2 + 16c^2 = 2(3ab + 6bc + 4ac)$$

$$\Rightarrow (2a)^2 + (3b)^2 + (4c)^2 - (2a)(3b)$$

$$- (3b)(4c) - (2a)(4c) = 0$$

$$\Rightarrow (2a-3b)^2 + 3b-4c)^2 + (4c-2a)^2 = 0$$

$$\Rightarrow (2a-3b)^2 = 0, (3b-4c)^2 = 0, (4c-2a)^2 = 0$$

$$\Rightarrow (2a-3b) = 0, (3b-4c) = 0, (4c-2a) = 0$$

$$\Rightarrow 2a = 3b = 4c$$

$$\Rightarrow \frac{a}{2} = \frac{b}{3} = \frac{c}{4}$$

Thus, $a, b, c \in \text{HP}$

126. It is given that a, b and c are in HP.

$$b = \frac{2ac}{a+c}$$

$$\begin{aligned}
\text{Now, } & \left(\frac{a+b}{2a-b} \right) + \left(\frac{c+b}{2c-b} \right) \\
&= \left(\frac{1+\frac{b}{a}}{2-\frac{b}{a}} \right) + \left(\frac{1+\frac{b}{c}}{2-\frac{b}{c}} \right) \\
&= \left(\frac{1+\frac{2c}{a+c}}{2-\frac{2c}{a+c}} \right) + \left(\frac{1+\frac{2a}{a+c}}{2-\frac{2a}{a+c}} \right) \\
&= \left(\frac{a+c+2c}{2a+2c-2c} \right) + \left(\frac{a+c+2a}{2a+2c-2a} \right) \\
&= \left(\frac{a+3c}{2a} \right) + \left(\frac{3a+c}{2c} \right) \\
&= \frac{ac+3c^2+3a^2+ac}{2ac} \\
&= \frac{3c^2+3a^2+2ac}{2ac} \\
&= \frac{3(a^2+c^2)}{2ac} + 1 \\
&= \frac{3}{2} \left(\frac{a}{c} + \frac{c}{a} \right) + 1 \\
&\geq \frac{3}{2} \times 2 + 1 = 4
\end{aligned}$$

Hence, the value of λ is 4.

127. As we know that,

AM $\geq$ GM

$$\Rightarrow \frac{\frac{a}{b} + \frac{b}{c} + \frac{c}{a}}{3} \geq \sqrt[3]{\frac{a}{b} \cdot \frac{b}{c} \cdot \frac{c}{a}} = 1$$

$$\Rightarrow \left(\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \right) \geq 3$$

Hence, the result.

128. It is given that a, b and c are in HP, so we can write,

$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in AP.

$$\text{Thus, } \frac{2}{b} = \frac{1}{a} + \frac{1}{c}$$

$$\begin{aligned}
\text{Now, } & \left(\frac{1}{b} + \frac{1}{c} - \frac{1}{a} \right) \left(\frac{1}{c} + \frac{1}{a} - \frac{1}{b} \right) \\
&= \left(\frac{3}{b} - \frac{1}{a} \right) \left(\frac{2}{b} - \frac{1}{b} \right) \\
&= \frac{1}{b} \left(\frac{3}{b} - \frac{1}{a} \right) = \frac{3}{b^2} - \frac{1}{ab}
\end{aligned}$$

129. Given a, b, c and d are in HP.

$$\Rightarrow \frac{1}{a}, \frac{1}{b}, \frac{1}{c}, \frac{1}{d} \text{ are in AP}$$

Let D be the common difference.

$$\text{Thus, } \frac{1}{b} - \frac{1}{a} = D$$

$$\Rightarrow ab = \frac{1}{D}(a-b)$$

$$\text{Similarly, } bc = \frac{1}{D}(b-c)$$

$$\text{and } cd = \frac{1}{D}(c-d)$$

Now, $ab + bc + cd$

$$= \frac{1}{D}(a-b) + \frac{1}{D}(b-c) + \frac{1}{D}(c-d)$$

$$= \frac{1}{D}(a-b+b-c+c-d)$$

$$= \frac{1}{D}(a-d)$$

$$= \left(\frac{3ad}{a-d} \right) \times (a-d)$$

$$= 3ad$$

130. Since a, b and c are in AP

$$\Rightarrow 2b = a + c \quad \dots(i)$$

Also, x, y and z are in HP

$$\Rightarrow y = \frac{2xz}{x+z} \quad \dots(ii)$$

And, ax, by and cz are in GP.

$$\therefore b^2y^2 = axcz \quad \dots(iii)$$

$$\Rightarrow \left(\frac{a+c}{2} \right)^2 \times \frac{4x^2z^2}{(x+z)^2} = axcz \text{ [from Eqs. (i) and (ii)]}$$

$$\Rightarrow \frac{(a+c)^2}{ac} = \frac{(x+z)^2}{xz}$$

$$\Rightarrow \frac{a^2+c^2+2ac}{ac} = \frac{x^2+z^2+2xz}{xz}$$

$$\Rightarrow \frac{a}{c} + \frac{c}{a} + 2 = \frac{x}{z} + \frac{z}{x} + 2$$

$$\Rightarrow \frac{a}{c} + \frac{c}{a} = \frac{x}{z} + \frac{z}{x}$$

Hence, the result.

131. We have,

$$\frac{a-x}{px} = \frac{a-y}{qy} = \frac{a-z}{rz}$$

$$\Rightarrow \frac{\frac{a}{x}-1}{p} = \frac{\frac{a}{y}-1}{q} = \frac{\frac{a}{z}-1}{r}$$

Since, p, q, r are in AP, so we can write,

$$2q = p + r$$

$$\Rightarrow 2\left(\frac{a}{y}-1\right) = \left(\frac{a}{x}-1\right) + \left(\frac{a}{z}-1\right)$$

$$\Rightarrow \frac{2a}{y} - 2 = \frac{a}{x} + \frac{a}{z} - 2$$

$$\Rightarrow \frac{2a}{y} = \frac{a}{x} + \frac{a}{z}$$

$$\Rightarrow \frac{1}{x}, \frac{1}{y}, \frac{1}{z} \text{ are in AP}$$

$$\Rightarrow x, y, z \text{ are in HP.}$$

132. Do yourself.

133. (i) Let $S_n = 1 + 2x + 3x^2 + 4x^3 + \dots + nx^{n-1}$

$$\Rightarrow x \cdot S_n = x + 2x^2 + 3x^3 + 4x^4 + \dots + nx^n$$

Subtracting, we get

$$\Rightarrow (1-x)S_n = 1 + x + x^2 + \dots + x^{n-1} - nx^n$$

$$= \left(\frac{1-x^n}{1-x}\right) - nx^n$$

$$\Rightarrow S_n = \left(\frac{1-x^n}{(1-x)^2}\right) - \frac{nx^n}{(1-x)}$$

134. Do yourself.

135. (i) Let $S_n = 1 + \frac{3}{2} + \frac{5}{4} + \frac{7}{8} + \dots + \frac{2n-1}{2^n}$

$$\Rightarrow \frac{S_n}{2} = \frac{1}{2} + \frac{3}{4} + \frac{5}{8} + \dots + \frac{2n-3}{2^n} + \frac{2n-1}{2^{n+1}}$$

Subtracting, we get

$$\left(1 - \frac{1}{2}\right)S_n = \left(1 + \frac{2}{2} + \frac{2}{4} + \frac{2}{8} + \dots + \frac{2}{2^n} - \frac{2n-1}{2^{n+1}}\right)$$

$$= -1 + 2\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n}\right) - \frac{2n-1}{2^{n+1}}$$

$$= -1 + 2\left(\frac{1 - \left(\frac{1}{2}\right)^{n+1}}{1 - \frac{1}{2}}\right) - \frac{2n-1}{2^{n+1}}$$

$$= -1 + 4\left(1 - \left(\frac{1}{2}\right)^{n+1}\right) - \frac{2n-1}{2^{n+1}}$$

$$= \left(3 - \left(\frac{1}{2}\right)^{n-1}\right) - \frac{2n-1}{2^{n+1}}$$

$$\Rightarrow S_n = \left(\frac{3}{2} - \left(\frac{1}{2}\right)^n\right) - \frac{2n-1}{2^{n+2}}$$

(ii) We have,

$$\frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{16} + \dots$$

$$= \left(1 - \frac{1}{2}\right) + \left(1 - \frac{1}{2^2}\right) + \left(1 - \frac{1}{2^3}\right) + \dots + \left(1 - \frac{1}{2^n}\right)$$

$$= n - \left(\frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^n}\right)$$

$$= n - \frac{1}{2} \left(\frac{1 - \left(\frac{1}{2}\right)^n}{1 - \frac{1}{2}}\right)$$

$$= n - \left(1 - \left(\frac{1}{2}\right)^n\right)$$

$$= (n + 2^n - 1)$$

(iii) We have,

$$\frac{2}{3} + \frac{8}{9} + \frac{26}{27} + \frac{80}{81} + \dots$$

$$= \left(1 - \frac{1}{3}\right) + \left(1 - \frac{1}{3^2}\right) + \left(1 - \frac{1}{3^3}\right) + \dots + \left(1 - \frac{1}{3^n}\right)$$

$$= n - \left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots + \frac{1}{3^n}\right)$$

$$= n - \frac{1}{3} \left(\frac{1 - \frac{1}{3^n}}{1 - \frac{1}{3}}\right)$$

$$= n - \frac{1}{2} \left(1 - \frac{1}{3^n}\right)$$

(iv) We have

$$2 + \frac{7}{4} + \frac{10}{8} + \frac{13}{16} + \frac{16}{32} + \dots$$

$$= \frac{4}{2} + \frac{7}{4} + \frac{10}{8} + \frac{13}{16} + \frac{16}{32} + \dots + \frac{3n+1}{2^n}$$

$$\text{Let } S = \frac{4}{2} + \frac{7}{4} + \frac{10}{8} + \frac{13}{16} + \frac{16}{32} + \dots + \frac{3n+1}{2^n}$$

$$\therefore \frac{S}{2} = \frac{4}{4} + \frac{7}{8} + \frac{10}{16} + \frac{13}{32} + \frac{16}{64} + \dots + \frac{3n+1}{2^{n+1}}$$

Subtracting, we get

$$\left(S - \frac{S}{2}\right) = 2 + \frac{3}{4} + \frac{3}{8} + \frac{3}{16} + \dots + \frac{3}{2^n} - \frac{3n+1}{2^{n+1}}$$

$$\begin{aligned} \Rightarrow \frac{S}{2} &= 2 + \frac{3}{4} + \frac{3}{8} + \frac{3}{16} + \dots + \frac{3}{2^n} - \frac{3n+1}{2^{n+1}} \\ \Rightarrow \frac{S}{2} &= 2 + \frac{3}{4} \left(1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-2}} \right) - \frac{3n+1}{2^{n+1}} \\ \Rightarrow \frac{S}{2} &= 2 + \frac{3}{2} \left(1 - \frac{1}{2^{n-1}} \right) - \frac{3n+1}{2^{n+1}} \\ \Rightarrow S &= 4 + 3 \left(1 - \frac{1}{2^{n-1}} \right) - \frac{3n+1}{2^n} \end{aligned}$$

136. (i) We have

$$\begin{aligned} 3 + (3+d)\frac{1}{4} + (3+2d)\frac{1}{4^2} + \dots \text{to } \infty &= \frac{44}{9} \\ \Rightarrow \frac{a}{1-r} + \frac{dr}{(1-r)^2} &= \frac{44}{9} \\ \Rightarrow \frac{3}{1-\frac{1}{4}} + \frac{d\left(\frac{1}{4}\right)}{\left(1-\frac{1}{4}\right)^2} &= \frac{44}{9} \\ \Rightarrow 4 + \frac{4}{9}d &= \frac{44}{9} \\ \Rightarrow \frac{4}{9}d &= \frac{44}{9} - 4 = \frac{8}{9} \\ \Rightarrow d &= 2 \end{aligned}$$

137 Let $t_r = r(r+1) = r^2 + r$.

$$\begin{aligned} \text{Then } S_n &= \sum_{r=1}^n t_r \\ &= \sum_{r=1}^n (r^2 + r) \\ &= \sum_{r=1}^n r^2 + \sum_{r=1}^n r \\ &= \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \\ &= \frac{n(n+1)}{2} \left(\frac{2n+1}{3} + 1 \right) \\ &= \frac{n(n+1)(n+2)}{3} \end{aligned}$$

138. Do yourself.

139. Do yourself.

140. Do yourself.

141. Let $t_n = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$.

$$\text{Then } S_n = \sum_{n=1}^n t_n$$

$$\begin{aligned} &= \frac{1}{2} \sum_{n=1}^n (n^2 + n) \\ &= \frac{1}{2} \left(\sum_{n=1}^n n^2 + \sum_{n=1}^n n \right) \\ &= \frac{1}{2} \left(\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \right) \\ &= \frac{n(n+1)}{4} \left(\frac{(2n+1)}{3} + 1 \right) \\ &= \frac{n(n+1)(n+2)}{6} \end{aligned}$$

142. Do yourself.

143. Let $t_n = (1^2 + 2^2 + 3^2 + \dots + n^2)$

$$\begin{aligned} &= \frac{n(n+1)(2n+1)}{6} \\ &= \frac{(2n^3 + 3n^2 + n)}{6} \\ &= \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6} \end{aligned}$$

$$\text{Then } S_n = \sum_{n=1}^n \left(\frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6} \right)$$

$$\begin{aligned} &= \frac{1}{3} \sum_{n=1}^n n^3 + \frac{1}{2} \sum_{n=1}^n n^2 + \frac{1}{6} \sum_{n=1}^n 1 \\ &= \frac{1}{3} \times \left(\frac{n(n+1)}{2} \right)^2 + \frac{1}{2} \times \frac{n(n+1)(2n+1)}{6} + \frac{1}{6} \times n \\ &= \frac{n(n+1)}{12} \times (1 + 2n + 1) + \frac{1}{6} \times n \\ &= \frac{n(n+1)^2}{6} + \frac{n}{6} \\ &= \frac{n}{6} (n^2 + 2n + 2). \end{aligned}$$

144. Let $t_n = \frac{1^2 + 2^2 + \dots + n^2}{1 + 2 + 3 + \dots + n}$

$$\begin{aligned} &= \frac{\frac{n(n+1)(2n+1)}{6}}{\frac{n(n+1)}{2}} \\ &= \frac{(2n+1)}{3} \end{aligned}$$

$$\begin{aligned} \text{Thus, } S_n &= \sum t_n = \sum \frac{(2n+1)}{3} \\ &= \frac{1}{3} (\sum 2n + \sum 1) \end{aligned}$$

$$= \frac{1}{3}(n(n+1) + n)$$

$$= \frac{(n^2 + 2n)}{3}$$

145. Let $t_n = \frac{1^3 + 2^3 + 3^3 + \dots + n^3}{1 + 3 + 5 + \dots + (2n-1)}$

$$= \frac{\left(\frac{n(n+1)}{2}\right)^2}{n^2}$$

$$= \frac{1}{4}(n^2 + 2n + 1)$$

Then $S_n = \sum_{n=1}^n \frac{1}{4}(n^2 + 2n + 1)$

$$= \frac{1}{4} \sum_{n=1}^n n^2 + \frac{1}{2} \sum_{n=1}^n n + \frac{1}{4} \sum_{n=1}^n 1$$

$$= \frac{1}{4} \times \frac{n(n+1)(2n+1)}{6} + \frac{1}{2} \times \frac{n(n+1)}{2} + \frac{1}{4} \times n$$

$$= \frac{n}{4} \left(\frac{n^2 + 3n + 1}{6} + n^2 + n + 1 \right)$$

$$= \left(\frac{n(7n^2 + 10n + 7)}{24} \right)$$

146. Let $S_n = \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots$

$$= \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n \cdot (n+1)}$$

$$= \frac{(2-1)}{1.2} + \frac{(3-2)}{2.3} + \frac{(4-3)}{3.4} + \dots + \frac{(n-1-n)}{n \cdot (n+1)}$$

$$= \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{n-1} - \frac{1}{n} \right)$$

$$= \left(1 - \frac{1}{n} \right)$$

$$= \left(\frac{n-1}{n} \right)$$

Note If $S_\infty = \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots \infty$, then $S_\infty = 1$.

147. Do yourself.

148. Let

$$S_n = \frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \dots$$

$$= \frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \dots + \frac{1}{n(n+1)(n+2)}$$

$$= \frac{1}{2} \left[\frac{(3-1)}{1.2.3} + \frac{(4-2)}{2.3.4} + \frac{(5-3)}{3.4.5} + \dots + \frac{(n+2-n)}{n(n+1)(n+2)} \right]$$

$$= \frac{1}{2} \left[\left(\frac{1}{1.2} - \frac{1}{2.3} \right) + \left(\frac{1}{2.3} - \frac{1}{3.4} \right) + \dots + \left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right) \right]$$

$$= \frac{1}{2} \left[\frac{1}{2} - \frac{1}{(n+1)(n+2)} \right]$$

Note If $S_\infty = \frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \dots$, then $S_\infty = 1/4$.

149. Do yourself.

150. Do yourself.

151. Let $t_n = n(n+1)^2$

$$= n(n^2 + 2n + 1)$$

$$= n^3 + 2n^2 + n$$

Then $S_n = \sum n^3 + 2\sum n^2 + \sum n$

$$= \left(\frac{n(n+1)}{2} \right)^2 + 2 \left(\frac{n(n+1)(2n+1)}{6} \right) + \frac{n(n+1)}{2}$$

$$= \frac{n(n+1)}{2} \left(\frac{n(n+1)}{2} + \frac{2(2n+1)}{3} + 1 \right)$$

$$= \frac{n(n+1)}{12} (3n^2 + 3n + 8n + 10)$$

$$= \frac{n(n+1)(3n^2 + 11n + 10)}{12}$$

$$= \frac{n(n+1)(n+2)(3n+5)}{12}$$

Also, let $T_n = n^2(n+1) = n^3 + n^2$

Then $S'_n = \sum n^3 + \sum n^2$

$$= \left(\frac{n(n+1)}{2} \right)^2 + \frac{n(n+1)(2n+1)}{6}$$

$$= \frac{n(n+1)}{2} \left(\frac{n(n+1)}{2} + \frac{2n+1}{3} \right)$$

$$= \frac{n(n+1)}{12} (3n^2 + 3n + 4n + 2)$$

$$= \frac{n(n+1)}{12} (3n^2 + 7n + 2)$$

$$= \frac{n(n+1)(n+2)(3n+1)}{12}$$

Thus, $\frac{S_n}{S'_n} = \frac{n(n+1)(n+2)(3n+5)}{n(n+1)(n+2)(3n+1)}$

$$= \frac{3n+5}{3n+1}$$

$$\begin{aligned}
 152. \text{ Let } t_n &= \frac{3n+1}{(n+1)(n+2)(n+3)} \\
 &= \frac{3n+3-2}{(n+1)(n+2)(n+3)} \\
 &= \frac{3}{(n+2)(n+3)} - \frac{2}{(n+1)(n+2)(n+3)} \\
 &= 3 \left(\frac{1}{(n+2)} - \frac{1}{(n+3)} \right) - \frac{(n+3) - (n+1)}{(n+1)(n+2)(n+3)} \\
 &= 3 \left(\frac{1}{(n+2)} - \frac{1}{(n+3)} \right) - \frac{1}{(n+1)(n+2)} \\
 &\quad + \frac{1}{(n+2)(n+3)} \\
 &= 3 \left(\frac{1}{n+2} - \frac{1}{n+3} \right) - \left(\frac{1}{n+1} - \frac{1}{n+2} \right) \\
 &\quad + \left(\frac{1}{n+2} - \frac{1}{n+3} \right) \\
 &= 4 \left(\frac{1}{n+2} - \frac{1}{n+3} \right) - \left(\frac{1}{n+1} - \frac{1}{n+2} \right)
 \end{aligned}$$

Thus,

$$\begin{aligned}
 S_n &= 4 \left(\left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) + \dots + \left(\frac{1}{n+2} - \frac{1}{n+3} \right) \right) \\
 &\quad - \left(\left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{n+1} - \frac{1}{n+2} \right) \right) \\
 &= 4 \left(\frac{1}{3} - \frac{1}{n+3} \right) - \left(\frac{1}{2} - \frac{1}{n+2} \right) \\
 &= \left(\frac{2}{3} - \frac{(3n+5)}{(n+2)(n+3)} \right)
 \end{aligned}$$

153. We have,

$$\begin{aligned}
 \text{Sum} &= \frac{1}{\log_2 4} + \frac{1}{\log_{2^2} 4} + \frac{1}{\log_{2^3} 4} + \dots + \frac{1}{\log_{2^n} 4} \\
 &= \log_4 2 + \log_4 (2^2) + \log_4 (2^3) + \dots + \log_4 (2^n) \\
 &= \log_4 (2 \cdot 2^2 \cdot 2^3 \cdot \dots \cdot 2^n) \\
 &= \log_4 (2^{1+2+3+\dots+n}) \\
 &= \log_{2^2} (2^{1+2+3+\dots+n}) \\
 &= \left(\frac{1+2+3+\dots+n}{2} \right) \log_2 (2) \\
 &= \left(\frac{1+2+3+\dots+n}{2} \right) \\
 &= \frac{n(n+1)}{4}
 \end{aligned}$$

154. We have,

$$\begin{aligned}
 &\sum_{i=1}^n \sum_{j=1}^i \sum_{k=1}^j 1 \\
 &= \sum_{i=1}^n \sum_{j=1}^i (j) \\
 &= \sum_{i=1}^n \left(\frac{n(n+1)}{2} \right) \\
 &= \frac{1}{2} \sum_{i=1}^n (n^2 + n) \\
 &= \frac{1}{2} \left(\sum_{i=1}^n n^2 + \sum_{i=1}^n n \right) \\
 &= \frac{1}{2} \left(\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \right) \\
 &= \frac{1}{2} \times \frac{n(n+1)}{2} \left(\frac{(2n+1)}{3} + 1 \right) \\
 &= \frac{n(n+1)(n+2)}{6}
 \end{aligned}$$

$$155. \text{ Let } S_n = 3 + 7 + 13 + 21 + \dots + T_{n-1} + T_n \quad \dots(i)$$

$$\text{Also, } S_n = 3 + 7 + 13 + 21 + \dots + T_{n-1} + T_n \quad \dots(ii)$$

Subtracting

(i) - (ii), we get

$$\begin{aligned}
 0 &= 3 + 4 + 6 + \dots + (T_n - T_{n-1}) - T_n \\
 \Rightarrow T_n &= 3 + 4 + 6 + \dots + (T_n - T_{n-1}) \\
 \Rightarrow T_n &= 3 + (4 + 6 + \dots + (T_n - T_{n-1})) \\
 &= 3 + \frac{n-1}{2} (2 \cdot 4 + (n-2)2) \\
 &= 3 + (n-1)(n+2) \\
 &= n^2 + n + 1
 \end{aligned}$$

$$\text{Therefore, } S_n = \sum (n^2 + n + 1)$$

$$\begin{aligned}
 &= \sum n^2 + \sum n + \sum 1 \\
 &= \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} + n \\
 &= \frac{n(n^2 + 3n + 5)}{3}
 \end{aligned}$$

156. Given series is

$$1 + 3 + 7 + 15 + 31 + \dots$$

Here, the differences between the successive terms are 2, 4, 8, 16, ..., which are in GP. Whenever the successive differences are in GP, we consider its n th term as

$$t_n = ar^n + br^{n-1} + c$$

In order to find the value of a , b and c , we put $n = 1, 2, 3$.

$$\text{Thus, } t_1 = 1 = a + b + c$$

$$t_2 = 3 = ar^2 + br + c,$$

$$\text{and } t_3 = 7 = ar^2 + br^2 + c$$

On solving, we get

$$r = 2, a = 0, b = 1, c = -1$$

Therefore, $t_n = 2^n - 1$.

$$\begin{aligned} \text{Hence, } S_n &= \sum 2^n - \sum 1 \\ &= \left(\frac{2^n - 1}{2 - 1} \right) - n \\ &= 2^n - n - 1 \end{aligned}$$

$$157. \text{ Let } S_n = 1 + 4 + 10 + 22 + \dots + T_{n-1} + T_n \quad \dots(i)$$

$$\text{Also } S_n = 1 + 4 + 10 + 22 + \dots + T_{n-1} + T_n \quad \dots(ii)$$

Subtracting

(i) - (ii), we get

$$\begin{aligned} 0 &= 1 + 3 + 6 + 12 + \dots + (T_n - T_{n-1}) - T_n \\ \Rightarrow T_n &= 1 + 3 + 6 + 12 + \dots + (T_n - T_{n-1}) \\ \Rightarrow T_n &= 1 + (3 + 6 + 12 + \dots + (T_n - T_{n-1})) \\ \Rightarrow T_n &= 1 + 3(1 + 2 + 2^2 + \dots + 2^{n-1}) \\ \Rightarrow T_n &= 1 + 3 \left(\frac{2^n - 1}{2 - 1} \right) = 3 \cdot 2^{n-1} - 2 \end{aligned}$$

$$\begin{aligned} \text{Therefore, } S_n &= \sum T_n \\ &= \sum (3 \cdot 2^{n-1} - 2) \\ &= 3 \sum 2^{n-1} - \sum 2 \\ &= 3 \left(\frac{2^n - 1}{2 - 1} \right) - 2n \\ &= 3 \cdot 2^n - 2n - 3 \end{aligned}$$

$$158. \text{ It is given that } x + y + z = 1$$

$$\therefore (1 - x) = y + z, (1 - y) = x + z, (1 - z) = x + y$$

$$\text{Solving, we have } x + y = \frac{1}{2}, y + z = \frac{1}{2} \text{ and } z + x = \frac{1}{2}$$

$$\text{We know that for any two numbers, } \frac{y+z}{2} \geq \sqrt{yz}$$

$$\Rightarrow (y+z) \geq 2\sqrt{yz}$$

$$\text{Similarly, } (x+z) \geq 2\sqrt{xz}$$

$$\text{and } (x+y) \geq 2\sqrt{xy}$$

On multiplication, we get,

$$(y+z)(z+x)(x+y) \geq 8xyz$$

$$(1-x)(1-y)(1-z) \geq 8xyz$$

$$\frac{xyz}{(1-x)(1-y)(1-z)} \leq \frac{1}{8}$$

Hence, the maximum value is $\frac{1}{8}$.

159. We have

$$\begin{aligned} &\left(p + \frac{1}{p} \right)^2 + \left(q + \frac{1}{q} \right)^2 \\ &= \left(p^2 + \frac{1}{p^2} \right) + \left(q^2 + \frac{1}{q^2} \right) + 4 \\ &= \left(p^2 + q^2 + \frac{1}{p^2} + \frac{1}{q^2} \right) + 4 \end{aligned}$$

$$\text{Now, } \left(\frac{p^2 + q^2}{2} \right) \geq \left(\frac{p+q}{2} \right)^2$$

$$\Rightarrow \left(\frac{p^2 + q^2}{2} \right) \geq \frac{1}{4}$$

$$\Rightarrow (p^2 + q^2) \geq \frac{1}{2} \quad \dots(i)$$

$$\text{Also, } \left(\frac{p^{-2} + q^{-2}}{2} \right) \geq \left(\frac{p+q}{2} \right)^{-2}$$

$$\Rightarrow \left(\frac{p^{-2} + q^{-2}}{2} \right) \geq 4$$

$$\Rightarrow \left(\frac{1}{p^2} + \frac{1}{q^2} \right) \geq 8$$

$$\text{Thus, } \left(p^2 + q^2 + \frac{1}{p^2} + \frac{1}{q^2} \right) + 4 \geq \frac{1}{2} + 8 + 4 = \frac{25}{2}$$

$$\Rightarrow \left(p + \frac{1}{p} \right)^2 + \left(q + \frac{1}{q} \right)^2 \geq \frac{25}{2}$$

Hence, the result.

160. We have,

$$\frac{\left(1 + \frac{1}{a} \right) + \left(1 + \frac{1}{b} \right) + \left(1 + \frac{1}{c} \right)}{3}$$

$$\geq \sqrt[3]{\left(1 + \frac{1}{a} \right) \cdot \left(1 + \frac{1}{b} \right) \cdot \left(1 + \frac{1}{c} \right)}$$

$$\Rightarrow \sqrt[3]{\left(1 + \frac{1}{a} \right) \cdot \left(1 + \frac{1}{b} \right) \cdot \left(1 + \frac{1}{c} \right)}$$

$$\leq \left(\frac{3 + \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)}{3} \right)$$

$$= \left(1 + \left(\frac{a^{-1} + b^{-1} + c^{-1}}{3} \right) \right)$$

$$\geq \left(1 + \left(\frac{a+b+c}{3} \right)^{-1} \right) = (1+3) = 4$$

$$\Rightarrow \sqrt[3]{\left(1 + \frac{1}{a} \right) \cdot \left(1 + \frac{1}{b} \right) \cdot \left(1 + \frac{1}{c} \right)} \geq 4$$

$$\Rightarrow \left(1 + \frac{1}{a}\right) \cdot \left(1 + \frac{1}{b}\right) \cdot \left(1 + \frac{1}{c}\right) \geq 4^3 = 64$$

161. As we know that,
AM $\geq$ HM

$$\therefore \left(\frac{a+b+c+d}{4}\right) \geq \frac{4}{\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}\right)}$$

$$\Rightarrow \left(\frac{a+b+c+d}{4}\right) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}\right) \geq 4$$

$$\Rightarrow (a+b+c+d) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}\right) \geq 16$$

163. As we know that,
AM $\geq$ GM

$$\therefore \left(\frac{a+b+c}{3}\right) \left(\frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{3}\right) \geq \sqrt[3]{(abc) \times \frac{1}{(abc)}}$$

$$\Rightarrow \left(\frac{a+b+c}{3}\right) \left(\frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{3}\right) \geq 1$$

$$\Rightarrow (a+b+c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq 9$$

$$\Rightarrow (a+b+c) \left(\frac{ab+bc+ca}{abc}\right) \geq 9$$

$$\Rightarrow (a+b+c)(ab+bc+ca) \geq 9abc$$

Hence, the result.

164. We have,

$$\sum_{r=0}^n (x^r) = (1+x)(1+x^2)(1+x^4)(1+x^8) \dots (1+x^{128})$$

$$= \left(\frac{1-x^{256}}{1-x}\right)$$

$$= \left(\frac{1-x^{256}}{1-x}\right)$$

$$\Rightarrow (1+x+x^2+x^3+\dots+x^n) = \left(\frac{1-x^{256}}{1-x}\right)$$

$$\Rightarrow \left(\frac{1-x^{256}}{1-x}\right) = \left(\frac{1-x^{n+1}}{1-x}\right)$$

$$\Rightarrow n+1 = 256$$

$$\Rightarrow n = 255$$

164 Do yourself.

165. Let the sides of a triangle be 2, c, $(a+\sqrt{b})$.

Since the sides of a triangle are in GP. So,

$$c^2 = 2(a+\sqrt{b}) \quad \dots(i)$$

$$\text{Also, } c^2 + 4 = (a+\sqrt{b})^2 \quad \dots(ii)$$

From Eqs (i) and (ii), we get

$$(a+\sqrt{b})^2 = 2(a+\sqrt{b}) + 4$$

$$\Rightarrow (a+\sqrt{b})^2 - 2(a+\sqrt{b}) = 4$$

$$\Rightarrow (a+\sqrt{b})^2 - 2(a+\sqrt{b}) + 1 = 4 + 1$$

$$\Rightarrow (a+\sqrt{b}-1)^2 = (\sqrt{5})^2$$

$$\Rightarrow (a+\sqrt{b}-1) = (\sqrt{5})$$

$$\Rightarrow (a+\sqrt{b}) = (1+\sqrt{5})$$

Thus, $a = 1$ and $b = 5$

Hence, the value of $a^2 + b^2 + 10 = 1 + 25 + 10 = 36$

166. Since a , b and c are the sides of a triangle, so,

$$a^2 + b^2 + c^2 > ab + bc + ca$$

$$\Rightarrow \frac{ab+bc+ca}{a^2+b^2+c^2} < 1 \quad \dots(i)$$

Also, $a < b+c$

$$\Rightarrow a^2 < a(b+c)$$

Similarly, $b^2 < b(a+c)$

and $c^2 < c(a+b)$

Thus, $a^2 + b^2 + c^2 < 2(ab+bc+ca)$

$$\Rightarrow 2(ab+bc+ca) > a^2 + b^2 + c^2$$

$$\Rightarrow \frac{(ab+bc+ca)}{(a^2+b^2+c^2)} > \frac{1}{2} \quad \dots(ii)$$

From Relations (i) and (ii), we get

$$\frac{1}{2} < \frac{(ab+bc+ca)}{(a^2+b^2+c^2)} < 1$$

167. Applying, AM $\geq$ GM, we have

$$\left(\frac{1+a}{2}\right) \geq \sqrt{a}$$

$$\Rightarrow (1+a) \geq 2\sqrt{a}$$

Similarly, $(1+b) \geq 2\sqrt{b}$

$$(1+c) \geq 2\sqrt{c}$$

$$\text{and } (1+d) \geq 2\sqrt{d}$$

Multiplying, we get

$$(1+a)(1+b)(1+c)(1+d) \geq 16\sqrt{abcd} = 16$$

Hence, $(1+a)(1+b)(1+c)(1+d) \geq 16$

168. Now,

$$\begin{aligned} (1+a)(1+b) - 2(ab+1) &= 1+a+b+ab-2ab-2 \\ &= -1+a+b-ab \\ &= -(1-a)(1-b) \\ &= -ve \end{aligned}$$

Thus, $(1+a)(1+b) < 2(ab+1)$

Similarly, $(1+c)(1+d) < 2(cd+1)$

Hence,

$$(1+a)(1+b)(1+c)(1+d) < 4(ab+1)(cd+1) < 8(abcd+1)$$

Hence, the result.

169. Applying AM $\geq$ GM, we get

$$\begin{aligned} \frac{(1-x) + (1-y) + (1-z)}{3} &\geq \sqrt[3]{(1-x)(1-y)(1-z)} \\ \Rightarrow \frac{3 - (x+y+z)}{3} &\geq \sqrt[3]{(1-x)(1-y)(1-z)} \\ \Rightarrow \frac{3-1}{3} &\geq \sqrt[3]{(1-x)(1-y)(1-z)} \\ \Rightarrow \frac{2}{3} &\geq \sqrt[3]{(1-x)(1-y)(1-z)} \\ \Rightarrow (1-x)(1-y)(1-z) &\leq \frac{8}{27} \end{aligned}$$

Hence, the result.

170. $\frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} = \frac{c+a+b}{abc} = \frac{1}{abc}$ (given)

We know that

$$\begin{aligned} \left(\frac{a+b+c}{3}\right) &\geq \sqrt[3]{abc} \\ \Rightarrow \left(\frac{a+b+c}{3}\right)^3 &\geq abc \\ \Rightarrow abc &\leq \frac{1}{27} \\ \Rightarrow \frac{1}{abc} &\geq 27 \\ \Rightarrow \left(\frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca}\right) &\geq 27 \end{aligned}$$

Hence, the result.

171. We have

$$\begin{aligned} (ax+by)^2 - (a^2+b^2)(x^2+y^2) &= a^2x^2 + b^2y^2 + 2abxy - a^2x^2 - a^2y^2 - b^2x^2 - b^2y^2 \\ &= -(a^2y^2 + b^2x^2 - 2abxy) \\ &= -(bc-ay)^2 \\ &= -ve \end{aligned}$$

Thus, $(ax+by)^2 < (a^2+b^2)(x^2+y^2)$

$$\Rightarrow (ax+by)^2 < 1 \cdot 1 = 1$$

$$\Rightarrow (ax+by) < 1$$

Hence, the result.

172. Applying AM $\geq$ GM, we get

$$\begin{aligned} \left(\frac{px+qy}{2}\right) &\geq \sqrt{pqxy} \\ \Rightarrow (px+qy) &\geq 2\sqrt{pqxy} \end{aligned}$$

Similarly, $(pq+xy) \geq 2\sqrt{pqxy}$

Thus, $(px+qy)(pq+xy) \geq 4pqxy$

Hence, the result.

173. We have,

$$\log_2 x + \log_2 y \geq 6$$

$$\Rightarrow \log_2(xy) \geq 6$$

$$\Rightarrow (xy) \geq 2^6 = 64$$

Applying AM $\geq$ GM, we have

$$\begin{aligned} \left(\frac{x+y}{2}\right) &\geq \sqrt{xy} \\ \Rightarrow \left(\frac{x+y}{2}\right) &\geq \sqrt{64} = 8 \\ \Rightarrow (x+y) &\geq 16 \end{aligned}$$

Hence, the minimum value of $(x+y)$ is 16.

174. Applying AM $\geq$ GM, we have

$$\begin{aligned} \left(\frac{\log_a b + \log_b a}{2}\right) &\geq \sqrt{\log_a b + \log_b a} \\ \Rightarrow \left(\frac{\log_a b + \log_b a}{2}\right) &\geq 1 \\ \Rightarrow \log_a b + \log_b a &\geq 2 \end{aligned}$$

Hence, the minimum value is 2.

175. We have,

$$\begin{aligned} 2 \log_{10} x - \log_x(.01) &= 2 \log_{10} x - \log_x(10)^{-2} \\ &= 2 \log_{10} x + 2 \log_x(10) \\ &= 2(\log_{10} x + \log_x(10)) \\ &\geq 2 \cdot 2 = 4 \end{aligned}$$

Hence, the least value of $2 \log_{10} x - \log_x(.01)$

is 4.

176. We know that,

$$\begin{aligned} \left(a^2 - \frac{1}{a^2}\right)^2 + \left(b^2 - \frac{1}{b^2}\right)^2 \\ = \left(a^4 + \frac{1}{a^4}\right) + \left(b^4 + \frac{1}{b^4}\right) - 4 \end{aligned}$$

$$\text{Also, } \left(\frac{(a^2)^2 + (b^2)^2}{2}\right) \geq \left(\frac{a^2 + b^2}{2}\right)^2$$

$$\Rightarrow \left(\frac{(a^2)^2 + (b^2)^2}{2}\right) \geq \frac{1}{4} \quad (\text{given})$$

$$\Rightarrow (a^4 + b^4) \geq \frac{1}{2}$$

$$\text{Also, } \frac{(a^2)^{-2} + (b^2)^{-2}}{2} \geq \left(\frac{a^2 + b^2}{2}\right)^{-2}$$

$$\Rightarrow \frac{(a^2)^{-2} + (b^2)^{-2}}{2} \geq 4$$

$$\Rightarrow \left(\frac{1}{a^4} + \frac{1}{b^4}\right) \geq 8$$

$$\text{Thus, } \left(a^4 + \frac{1}{a^4} + b^4 + \frac{1}{b^4} - 4\right) \geq \frac{1}{2} + 8 - 4$$

$$\Rightarrow \left(a^2 - \frac{1}{a^2}\right)^2 + \left(b^2 - \frac{1}{b^2}\right)^2 \geq \frac{9}{2}$$

Hence, the result

177. We have

$$(m-a)(m-b)(m-c)(m-d) \\ = (b+c+d)(a+c+d)(a+b+d)(a+b+c)$$

Applying AM $\geq$ GM, we get

$$\left(\frac{b+c+d}{3}\right) \geq \sqrt[3]{bcd}$$

$$\Rightarrow \left(\frac{a+c+d}{3}\right) \geq \sqrt[3]{acd}$$

$$\Rightarrow \left(\frac{a+b+d}{3}\right) \geq \sqrt[3]{abd}$$

$$\text{Similarly, } \left(\frac{a+b+c}{3}\right) \geq \sqrt[3]{abc}$$

On multiplication, we get

$$(b+c+d)(a+c+d)(a+b+d)(a+b+c) \\ > 81abcd$$

Hence, the result.

178. We know that,

$$\left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 = (a^2 + b^2) + \left(\frac{1}{a^2} + \frac{1}{b^2}\right) + 4$$

$$\text{Now, } \left(\frac{a^2 + b^2}{2}\right) \geq \left(\frac{a+b}{2}\right)^2$$

$$\Rightarrow a^2 + b^2 \geq \frac{1}{2}$$

$$\text{Also, } \left(\frac{(a)^{-2} + (b)^{-2}}{2}\right) \geq \left(\frac{a+b}{2}\right)^{-2}$$

$$\Rightarrow \left(\frac{1}{a^2} + \frac{1}{b^2}\right) \geq 8 \quad (\because \text{given } a+b=1)$$

Thus,

$$(a^2 + b^2) + \left(\frac{1}{a^2} + \frac{1}{b^2}\right) + 4 \geq \frac{1}{2} + 8 + 4 = \frac{25}{2}$$

$$\text{Hence, } \left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 \geq \frac{25}{2}$$

179. We have $(ax + by)^2 - (a^2 + b^2)(x^2 + y^2)$

$$= a^2x^2 + b^2y^2 + 2abxy - a^2x^2 - a^2y^2 - b^2x^2 - b^2y^2$$

$$= 2abxy - a^2y^2 - b^2x^2$$

$$= -(a^2y^2 + b^2x^2 - 2abxy)$$

$$= -(ay - bx)^2 < 0$$

$$\text{Thus, } (ax + by)^2 < (a^2 + b^2)(x^2 + y^2) = 16$$

$$\Rightarrow (ax + by) < 4$$

Hence, the greatest value of $(ax + by)$ is 4.

180. We know that

$$\left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 + \left(c + \frac{1}{c}\right)^2$$

$$(a^2 + b^2 + c^2) + \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right) + 6$$

$$\text{and } \left(\frac{a^2 + b^2 + c^2}{3}\right) \geq \left(\frac{a+b+c}{3}\right)^2$$

$$\Rightarrow \left(\frac{a^2 + b^2 + c^2}{3}\right) \geq \frac{1}{9} \quad (\text{given})$$

$$\Rightarrow (a^2 + b^2 + c^2) \geq \frac{1}{3}$$

$$\text{Also, } \left(\frac{(a)^{-2} + (b)^{-2} + (c)^{-2}}{3}\right) \geq \left(\frac{a+b+c}{3}\right)^{-2}$$

$$\Rightarrow \left(\frac{(a)^{-2} + (b)^{-2} + (c)^{-2}}{3}\right) \geq 9$$

$$\Rightarrow \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right) \geq 27$$

Thus,

$$(a^2 + b^2 + c^2) + \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right) + 6 \geq \frac{1}{3} + 27 + 6$$

$$\left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 + \left(c + \frac{1}{c}\right)^2 \geq \frac{100}{3}$$

Hence, the least value is $\frac{100}{3}$.

181. It is given that

$$a + 2b + 3c = 12$$

Applying, AM $\geq$ GM, we have

$$\frac{a + b + b + c + c + c}{6} \geq \sqrt[6]{ab^2c^3}$$

$$\Rightarrow \frac{a + 2b + 3c}{6} \geq \sqrt[6]{ab^2c^3}$$

$$\Rightarrow \frac{12}{6} \geq \sqrt[6]{ab^2c^3}$$

$$\Rightarrow (ab^2c^3) \leq 2^6 = 64$$

Hence, the maximum value is 64.

182. It is given that $4a + 3b + 2c = 45$

Applying AM $\geq$ GM, we have

$$\left(\frac{2a + 2a + b + b + b + \frac{c}{2} + \frac{c}{2} + \frac{c}{2} + \frac{c}{2}}{9}\right)$$

$$\geq \sqrt[9]{a^2b^3c^4}$$

$$\Rightarrow \left(\frac{4a + 3b + 2c}{9}\right) \geq \sqrt[9]{a^2b^3c^4}$$

$$\Rightarrow \left(\frac{45}{9}\right) \geq \sqrt[9]{\frac{a^2 b^3 c^4}{4}}$$

$$\Rightarrow (a^2 b^3 c^4) \leq 4(5)^9$$

Hence, the greatest value of $(a^2 b^3 c^4)$ is $4(5)^9$.

183. Given $2x + 3y = 10$

Now $2x + 3y = x + x + y + y + y$

Applying AM $\geq$ GM, we have

$$\left(\frac{x+x+y+y+y}{5}\right) \geq \sqrt[5]{x \cdot x \cdot y \cdot y \cdot y}$$

$$\Rightarrow \left(\frac{2x+3y}{5}\right) \geq \sqrt[5]{x^2 y^3}$$

$$\Rightarrow \left(\frac{10}{5}\right) \geq \sqrt[5]{x^2 y^3}$$

$$\Rightarrow (2)^5 \geq x^2 y^3$$

$$\Rightarrow x^2 y^3 \leq 2^5 = 32$$

Hence, the maximum value of $x^2 y^3$ is 32.

184. It is given that $x + 5y = 18$.

Applying AM $\geq$ GM, we have

$$\Rightarrow \left(\frac{x+y+y+y+y+y}{6}\right) \geq \sqrt[6]{xy^5}$$

$$\Rightarrow \left(\frac{x+5y}{6}\right) \geq \sqrt[6]{xy^5}$$

$$\Rightarrow \left(\frac{18}{6}\right) \geq \sqrt[6]{xy^5}$$

$$\Rightarrow (xy^5) \leq 3^6 = 729$$

Hence, the maximum value of (xy^5) is 729.

185. It is given that a, b and c are in AP.

$$\Rightarrow 2b = a + c$$

Applying, AM $\geq$ GM

$$\Rightarrow \left(\frac{a+b+c}{3}\right) \geq \sqrt[3]{abc}$$

$$\Rightarrow \left(\frac{3b}{3}\right) \geq \sqrt[3]{abc}$$

$$\Rightarrow b \geq \sqrt[3]{64} = 4$$

$$\Rightarrow b \geq 4$$

Hence, the minimum value of b is 4.

186. We have

$$\begin{aligned} f(x) &= (x^2 + 2) + \frac{1}{(x^2 + 1)} \\ &= (x^2 + 1) + \frac{1}{(x^2 + 1)} + 1 \\ &\geq 2 + 1 = 3 \end{aligned}$$

Hence, the minimum value of $f(x) = x^2 + 2 + \frac{1}{x^2 + 1}$ is 3.

187. Clearly, the minimum value of $f(x) = x^{10} + \frac{10}{x}$, $x > 0$ is 11.

188. Clearly, the min. value of $f(x) = x^{2012} + \frac{2012}{x}$, $x > 0$ is 2013.

189. We have

$$\begin{aligned} &\left(a^9 + \frac{1}{a^9}\right) + \left(a^7 + \frac{1}{a^7}\right) + \left(a^5 + \frac{1}{a^5}\right) \\ &+ \left(a^3 + \frac{1}{a^3}\right) + 1 \\ &\geq 2 + 2 + 2 + 2 + 1 = 9 \end{aligned}$$

Hence, the minimum value is 9.

190. We have

$$\begin{aligned} 2^{x^2+2x+2} + 2^{2-2x-x^2} &= 4 \cdot 2^{x^2+2x} + \frac{4}{2^{x^2+2x}} \\ &= 4 \cdot \left(2^{x^2+2x} + \frac{1}{2^{x^2+2x}}\right) \\ &\geq 4 \cdot 2 = 8 \end{aligned}$$

Hence, the minimum value is 8.

191. We have

$$\begin{aligned} (2^x + 2^{-x}) + (3^x + 3^{-x}) + (4^x + 4^{-x}) + (5^x + 5^{-x}) + 10 \\ \geq 2 + 2 + 2 + 2 + 10 = 18 \end{aligned}$$

Hence, the minimum value is 18.

192. It is given that,

$$a + 2b + 3c + 4d = 10$$

We know that

$$\begin{aligned} \frac{(a+2b) + (3c+4d)}{2} &\geq \sqrt{(a+2b)(3c+4d)} \\ \Rightarrow \frac{10}{2} &\geq \sqrt{(a+2b)(3c+4d)} \\ (a+2b)(3c+4d) &\leq 25 \end{aligned}$$

Thus, $M = 25$

$$\begin{aligned} \text{Also, } \frac{(a+c+2d) + 2(b+c+d)}{2} \\ \geq \sqrt{2(a+c+2d)(b+c+2d)} \end{aligned}$$

$$\frac{10}{2} \geq \sqrt{2(a+c+2d)(b+c+2d)}$$

$$(a+c+2d)(b+c+2d) \leq \frac{25}{2}$$

$$\text{Thus, } N = \frac{25}{2}$$

$$\begin{aligned} \text{Therefore, } M + 2N + 10 \\ = 25 + 25 + 10 = 60 \end{aligned}$$

193. We know that

$$\begin{aligned} \left(\frac{a^{-2}+b^{-2}+c^{-2}}{3}\right) &\geq \left(\frac{a+b+c}{3}\right)^{-2} \\ \Rightarrow \left(\frac{a^{-2}+b^{-2}+c^{-2}}{3}\right) &\geq 9 \quad (\text{given}) \\ \Rightarrow \left(\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}\right) &\geq 27 \end{aligned}$$

Hence, the minimum value is 27.

194. We know that

$$\begin{aligned} \left(\frac{x^3+y^3+z^3}{3}\right) &\geq \left(\frac{x+y+z}{3}\right)^3 \\ \Rightarrow \left(\frac{x^3+y^3+z^3}{3}\right) &\geq \frac{1}{27} \\ \Rightarrow (x^3+y^3+z^3) &\geq \frac{1}{9} \end{aligned}$$

$$\text{Also, } \left(\frac{x^6+y^6+z^6}{3}\right) \geq \left(\frac{x+y+z}{3}\right)^6$$

$$\Rightarrow \left(\frac{x^6+y^6+z^6}{3}\right) \geq \frac{1}{729}$$

$$\Rightarrow (x^6+y^6+z^6) \geq \frac{1}{243}$$

Thus, the least value of $(x^3+y^3+z^3)(x^6+y^6+z^6)$ is $\frac{1}{2187}$.

195. Applying AM $\geq$ GM, we have

$$\left(\frac{1+a+a^2}{3}\right) \geq \sqrt[3]{a^3} = a$$

$$\Rightarrow (1+a+a^2) \geq 3a$$

$$\text{Similarly, } (1+b+b^2) \geq 3b$$

$$\text{and } (1+c+c^2) \geq 3c$$

$$\text{Thus, } (1+a+a^2)(1+b+b^2)(1+c+c^2) \geq 27abc$$

$$\Rightarrow (1+a+a^2)(1+b+b^2)(1+c+c^2) \geq 27 \times 8$$

$$\Rightarrow (1+a+a^2)(1+b+b^2)(1+c+c^2) \geq 216$$

Hence, the least value is 216.

196. We know that,

$$\left(\frac{1+x^3}{2}\right) \geq \sqrt{x^3}$$

$$\Rightarrow (1+x^3) \geq 2\sqrt{x^3}$$

$$\text{Similarly, } (1+y^3) \geq 2\sqrt{y^3}$$

$$(1+z^3) \geq 2\sqrt{z^3}$$

$$\text{Thus, } (1+x^3)(1+y^3)(1+z^3) \geq 2\sqrt{(xyz)^3}$$

$$\Rightarrow (1+x^3)(1+y^3)(1+z^3) \geq 2\sqrt{8} = 4\sqrt{2}$$

Hence, the minimum value is $4\sqrt{2}$.

197. We know that,

$$\left(1+\frac{a}{2}+\frac{a}{2}\right) \geq 3 \times \sqrt[3]{1 \cdot \frac{a}{2} \cdot \frac{a}{2}}$$

$$\Rightarrow (1+a) \geq 3\sqrt[3]{\frac{a^2}{4}}$$

$$\text{Similarly, } (1+b) \geq 3\sqrt[3]{\frac{b^2}{4}}$$

On multiplication, we get

$$\{(1+a)(1+b)\}^3 \geq \frac{3^6}{16}(a^2b^2)$$

198. We know that,

$$(1+a)(1+b)(1+c)$$

$$= 1 + (a+b+c) + (ab+bc+ca) + abc$$

Now, applying AM $\geq$ GM

$$\Rightarrow \frac{(a+b+c) + (ab+bc+ca) + abc}{7} \geq (a^4b^4c^4)^{1/7}$$

$$\Rightarrow (a+b+c) + (ab+bc+ca) + abc \geq 7(a^4b^4c^4)^{1/7}$$

$$\begin{aligned} \text{Thus, } 1 + (a+b+c) + (ab+bc+ca) + abc \\ > (a+b+c) + (ab+bc+ca) + abc \\ &\geq 7(a^4b^4c^4)^{1/7} \end{aligned}$$

Hence, the result.

199. We know that,

$$\frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} = \frac{a+b+c}{abc} = \frac{1}{abc}$$

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc}$$

$$\Rightarrow \frac{1}{3} \geq \sqrt[3]{abc}$$

$$\Rightarrow \frac{1}{27} \geq abc$$

$$\Rightarrow abc \leq \frac{1}{27}$$

$$\Rightarrow \frac{1}{abc} \geq 27$$

$$\text{Hence, } \frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} \geq 27.$$

200. As we know, by Tchebycheff's Inequality

$$\left(\frac{a+b+c+d}{4}\right)^2 \leq \frac{a^2+b^2+c^2+d^2}{4}$$

$$\Rightarrow \left(\frac{8-e}{4}\right)^2 \leq \frac{16-e^2}{4} \quad (\because a^2+b^2+c^2+d^2+e^2=16)$$

$$\Rightarrow 64 + e^2 - 16e \leq 4(16 - e^2)$$

$$\Rightarrow 5e^2 - 16e \leq 0$$

$$\Rightarrow e(5e - 16) \leq 0$$

$$\therefore 0 \leq e \leq \frac{5}{16}$$

Thus, the range of e is $\left[0, \frac{5}{16}\right]$.

201. As we know that,

AM $\geq$ GM

$$\Rightarrow \frac{ab + bc + ca}{3} \geq \sqrt[3]{(ab)(bc)(ca)}$$

$$\Rightarrow \frac{8}{3} \geq \sqrt[3]{(ab)(bc)(ca)}$$

$$\Rightarrow \left(\frac{8}{3}\right)^3 \geq (abc)^2$$

$$\Rightarrow (abc)^2 \leq \left(\frac{8}{3}\right)^3$$

$$\Rightarrow (abc) \leq \left(\frac{8}{3}\right)^{3/2}$$

Hence, the maximum value of abc is $\left(\frac{8}{3}\right)^{3/2}$.

202. Given equation is

$$ax^4 + bx^3 + cx^2 + dx + 6 = 0$$

It is given that α, β, γ and δ be its roots.

Thus, $\alpha + \beta + \gamma + \delta = -b$

$$\sum \alpha\beta = c$$

$$\sum \alpha\beta\gamma = -d$$

and $\sum \alpha\beta\gamma\delta = 6$

Applying AM $\geq$ GM, we have,

$$\left(\frac{\alpha + \beta + \gamma + \delta}{4}\right) \left(\frac{\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta}{4}\right) \geq \sqrt[4]{\alpha\beta\gamma\delta} \sqrt[4]{\alpha^3\beta^3\gamma^3\delta^3} = \alpha\beta\gamma\delta = 6$$

$$\Rightarrow \left(-\frac{b}{4}\right) \left(-\frac{d}{4}\right) \geq 6$$

$$\Rightarrow bd \geq 96$$

Hence, the minimum value of bd is 96.

203. As we know that AM $\geq$ GM,

where $a + 2b + c = a + b + b + c$

$$\text{Thus, } \left(\frac{(a+b) + (b+c)}{2}\right) \geq \sqrt{(a+b)(b+c)}$$

$$\Rightarrow (a+b)(b+c) \leq \left(\frac{4}{2}\right)^2 = 4$$

$$\Rightarrow (ab + bc + ca + b^2) \leq 4$$

$$\Rightarrow (ab + bc + ca) \leq 4 - b^2$$

Hence the maximum value of $(ab + bc + ca)$ is 4.

204. As we know that AM $\geq$ GM

$$\text{Thus, } \left(\frac{a+b+c+d}{2}\right) \geq \sqrt{(a+b)(c+d)}$$

$$\text{Hence, } (a+b)(c+d) \leq \left(\frac{2}{2}\right)^2 = 1$$

Thus the maximum value of $(a+b)(c+d)$ is 1.

$$205. \left(\frac{1}{a} - 1\right) = \left(\frac{1-a}{a}\right)$$

$$= \left(\frac{a+b+c-a}{a}\right)$$

$$= \left(\frac{b+c}{a}\right)$$

$$\text{Similarly } \frac{1}{b} - 1 = \frac{c+a}{b} \text{ and } \frac{1}{c} - 1 = \frac{a+b}{c}$$

multiplying, we get

$$\left(\frac{1}{a} - 1\right) \left(\frac{1}{b} - 1\right) \left(\frac{1}{c} - 1\right) =$$

$$\left(\frac{b+c}{a}\right) \left(\frac{c+a}{b}\right) \left(\frac{a+b}{c}\right) \geq \left(\frac{8abc}{abc}\right) = 8$$

Hence the minimum value is 8.

206. As we know that AM $\geq$ HM

$$\text{Thus, } \left(\frac{(b+c) + (c+a) + (a+b)}{3}\right)$$

$$\geq \left(\frac{3}{\frac{1}{(b+c)} + \frac{1}{(c+a)} + \frac{1}{(a+b)}}\right)$$

$$\Rightarrow \frac{2}{3}(a+b+c) \left(\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b}\right) \geq 3$$

$$\Rightarrow (a+b+c) \left(\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b}\right) \geq \frac{9}{2}$$

$$\Rightarrow \left(\frac{a+b+c}{b+c} + \frac{a+b+c}{c+a} + \frac{a+b+c}{a+b}\right) \geq \frac{9}{2}$$

$$\Rightarrow \left(\frac{a}{b+c} + 1 + \frac{b}{c+a} + 1 + \frac{c}{a+b} + 1\right) \geq \frac{9}{2}$$

$$\Rightarrow \left(\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}\right) \geq \frac{9}{2} - 3 = \frac{3}{2}$$

Hence, the result.

$$\begin{aligned} 207. (1+x)(1+y) - 2(xy+1) \\ &= 1 + x + y + xy - 2xy - 2 \\ &= -1 + x + y - xy \\ &= -(x-1)(y-1) \\ &= -ve, \text{ since } x > 1, y > 1 \end{aligned}$$

Thus, $(1+x)(1+y) < 2(xy+1)$

Similarly, $(1+z)(1+w) < 2(zw+1)$

Therefore,

$$(1+x)(1+y)(1+z)(1+w) < 4(xy+1)(zw+1)$$

208. It can be easily proved that

$$(1+x)(1+y)(1+z)(1+w) < 4(xy+1)(zw+1)$$

Obviously, $(1+xy)(1+zw) < 2(xyzw+1)$

Hence, $(1+x)(1+y)(1+z)(1+w) < 8(xyzw+1)$

209. As we know that $AM \geq GM$

$$\text{or } \frac{a^4+b^4}{2} \geq \sqrt{a^4b^4}$$

$$\Rightarrow (a^4+b^4) \geq 2a^2b^2$$

$$\Rightarrow (a^4+b^4+c^2) \geq 2a^2b^2+c^2$$

$$\Rightarrow (a^4+b^4+c^2) \geq 2a^2b^2+c^2 \geq 2\sqrt{2a^2b^2c^2}$$

$$\Rightarrow (a^4+b^4+c^2) \geq 2\sqrt{2}abc$$

Hence, the result.

210. Since,

$$\frac{1}{2}\left(\frac{ab}{c} + \frac{bc}{a}\right) \geq \sqrt{\frac{ab}{c} \times \frac{bc}{a}} = b \quad \dots(i)$$

$$\frac{1}{2}\left(\frac{bc}{a} + \frac{ca}{b}\right) \geq \sqrt{\frac{bc}{a} \times \frac{ca}{b}} = c \quad \dots(ii)$$

$$\frac{1}{2}\left(\frac{ca}{b} + \frac{ab}{c}\right) \geq \sqrt{\frac{ca}{b} \times \frac{ab}{c}} = a \quad \dots(iii)$$

Adding relations (i), (ii) and (iii), we get

$$\frac{ab}{c} + \frac{bc}{a} + \frac{ca}{b} \geq a+b+c$$

Hence, the result.

211. By the m th power theorem,

$$\left(\frac{b^2+c^2}{2}\right) \geq \left(\frac{b+c}{2}\right)^2$$

$$\Rightarrow \left(\frac{b^2+c^2}{b+c}\right) \geq \left(\frac{b+c}{2}\right) \quad \dots(i)$$

$$\text{similarly, } \left(\frac{c^2+a^2}{c+a}\right) \geq \left(\frac{c+a}{2}\right) \quad \dots(ii)$$

$$\text{and } \left(\frac{a^2+b^2}{a+b}\right) \geq \left(\frac{a+b}{2}\right) \quad \dots(iii)$$

Adding relations (i), (ii) and (iii), we get

$$\left(\frac{b^2+c^2}{b+c} + \frac{c^2+a^2}{c+a} + \frac{a^2+b^2}{a+b}\right) \geq (a+b+c)$$

Hence, the result.

212. By the m th power theorem, we get

$$\left(\frac{x^{-2}+y^{-2}+z^{-2}}{3}\right) \geq \left(\frac{x^2+y^2+z^2}{3}\right)^{-1}$$

$$\Rightarrow \left(\frac{x^{-2}+y^{-2}+z^{-2}}{3}\right) \geq \left(\frac{3}{x^2+y^2+z^2}\right)$$

$$\Rightarrow x^{-2}+y^{-2}+z^{-2} \geq \left(\frac{9}{1}\right) = 9$$

$$\Rightarrow \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} \geq 9$$

$$213. a^{-5} + a^{-4} + 3a^{-3} + 1 + a^8 + a^{10}$$

$$= a^{-5} + a^{-4} + a^{-3} + a^{-3} + a^{-3} + 1 + a^8 + a^{10}$$

$$= \left(\frac{a^{-5} + a^{-4} + a^{-3} + a^{-3} + a^{-3} + 1 + a^8 + a^{10}}{9} \right)$$

$$\geq \sqrt[9]{a^{-5} \cdot a^{-4} \cdot a^{-3} \cdot a^{-3} \cdot a^{-3} \cdot a^8 \cdot a^{10}} = 1$$

$$\Rightarrow \left(\frac{a^{-5} + a^{-4} + a^{-3} + a^{-3} + a^{-3} + 1 + a^8 + a^{10}}{9} \right) > 1$$

$$\Rightarrow (a^{-5} + a^{-4} + a^{-3} + a^{-3} + a^{-3} + 1 + a^8 + a^{10}) > 1$$

Hence, the minimum value is 1.